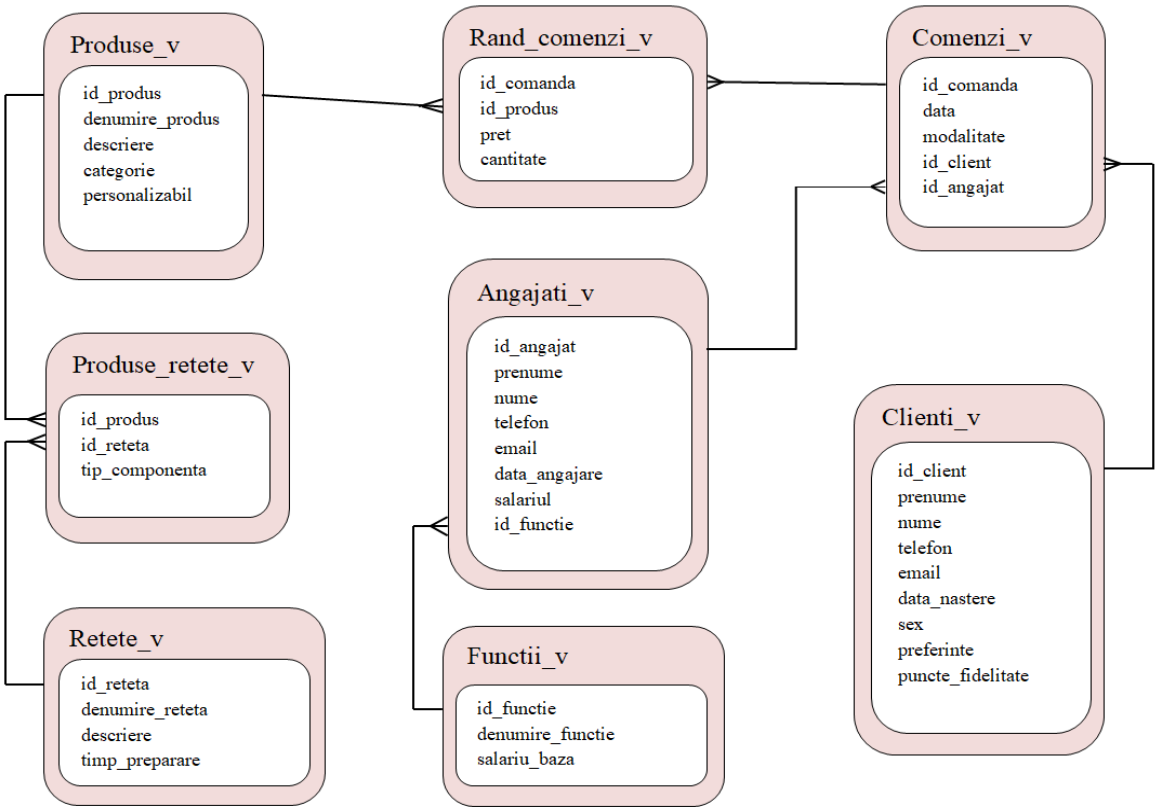
PROIECT ORACLE

1. Descrierea bazei de date: obiectivul proiectului, descrierea tabelelor si a atributelor, precizarea restrictiilor si a tipurilor de legaturi (max. ½ pag.). BD trebuie sa fie normalizata cel putin in FN3.

Baza de date urmărește gestionarea activității cofetăriei Vera, incluzând informații despre clienți, angajați, produse, comenzi, rețete, ingrediente și furnizori, pentru a organiza eficient afacerea.

1) **Angajati_v:** id_angajat (primary key), prenume, nume, email, telefon, data_angajare, salariul, id_functie (foreign key). Relație 1:M cu tabela *Funcții*.
2) **Functii_v:** id_functie (primary key), denumire_functie, salariu_baza.
3) **Clienti_v:** id_client (primary key), prenume, nume, telefon, email, data_nastere, sex (M/F), preferinte, puncte_fidelitate. Relație 1:M cu *Comenzi*.
4) **Comenzi_v:** id_comanda (primary key), data, modalitate, id_client (foreign key), id_angajat (foreign key). Relație 1:M cu *Rand_comenzi*.
5) **Rand_comenzi_v:** id_comanda, id_produs, pret, cantitate. Este o tabelă de legatură între *Comenzi* și *Produse*.
6) **Produse_v:** id_produs (primary key), denumire_produs, descriere, categorie, personalizabil. Relație 1:M cu *Rand_comenzi*.
7) **Produse_retete_v**: id_produs, id_reteta, tip_componenta. Este o tabelă de legatură între *Rețetei* și *Produse*.
8) **Retete_v:** id_reteta (primary key), denumire_reteta, descriere, timp_preparare. Relație 1:M cu *Produse.*

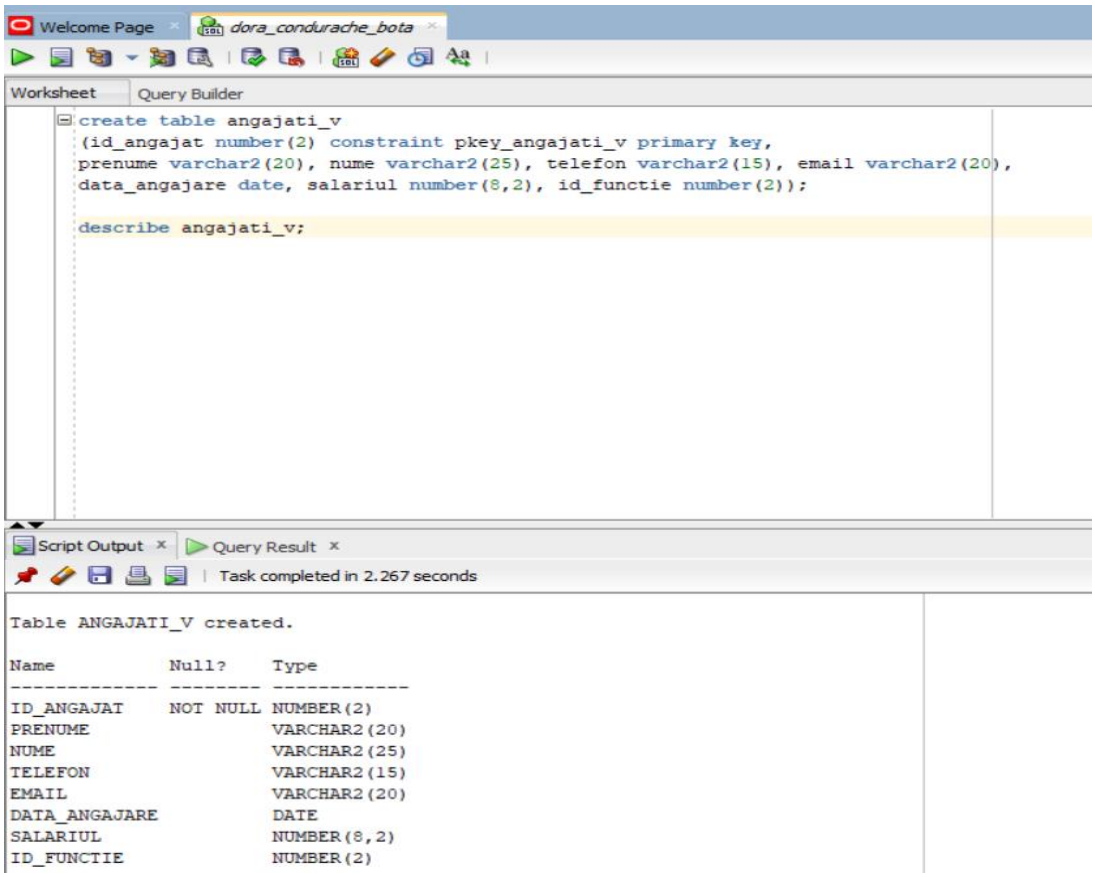
2. Schema corectă a bazei de date



3. Crearea tabelelor (min. 4 tabele) și adăugarea (min 10, max 20) de înregistrări în fiecare tabelă

Tabela Angajati_v:

create table angajati_v
(id_angajat number(2) constraint pkey_angajati_v primary key,
prenume varchar2(20), nume varchar2(25), telefon varchar2(15), email varchar2(20),
data_angajare date, salariul number(8,2), id_functie number(2));



alter table angajati_v
add constraint fkey_angajati_v foreign key (id_functie)
references functii_v (id_functie);

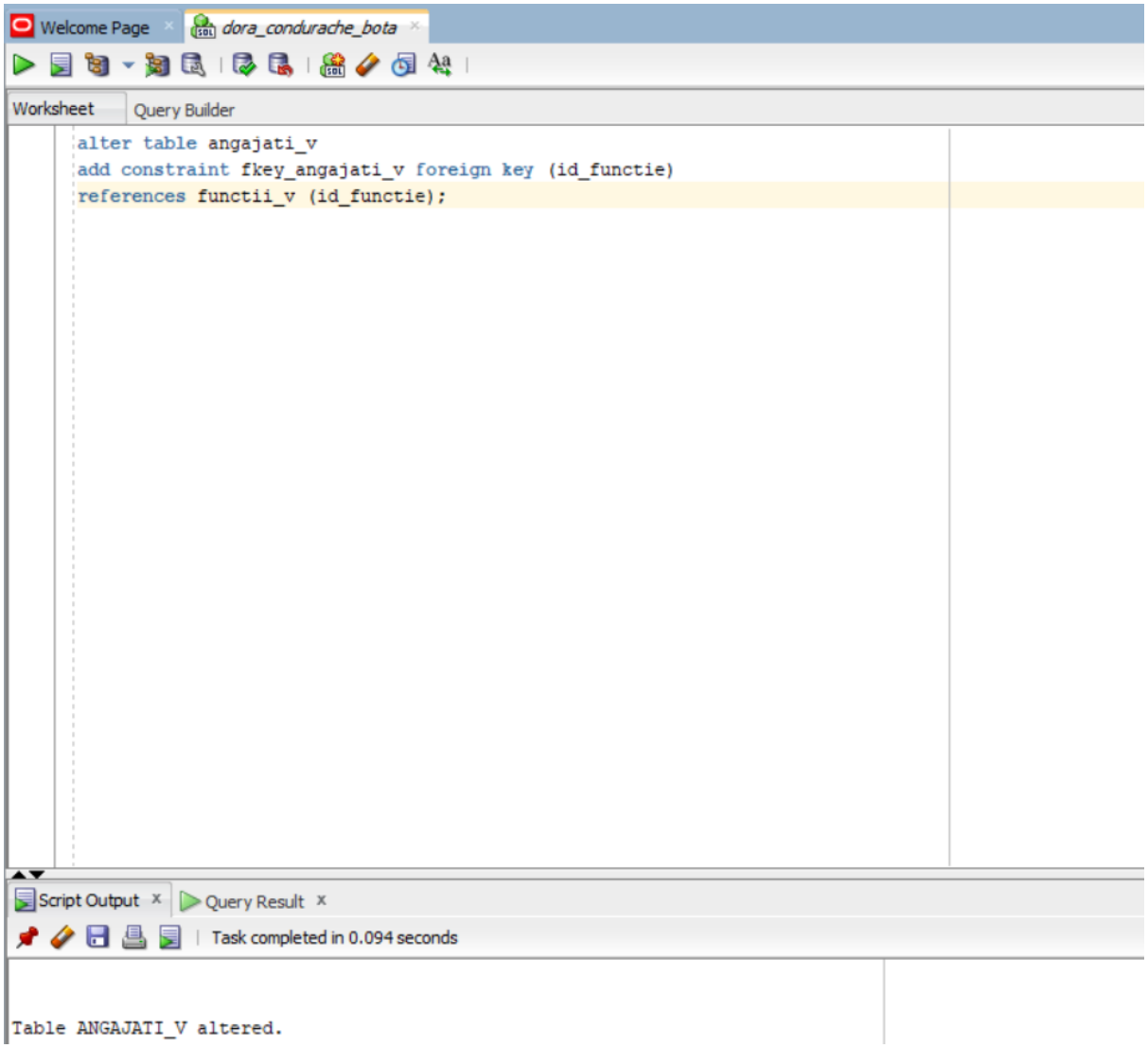

Welcome Page
dora_condurache_bota
Worksheet
Query Builder
alter table angajati_v
add constraint fkey_angajati_v foreign key (id_functie)
references functii_v (id_functie);
Script Output
Query Result
Task completed in 0.094 seconds
Table ANGAJATI_V altered.

Tabela Functii_v:

```
create table functii_v
(id_functie number(2) constraint pkey_functii_v primary key,
denumire_functie varchar2(30), salariu_baza number(8,2));
```

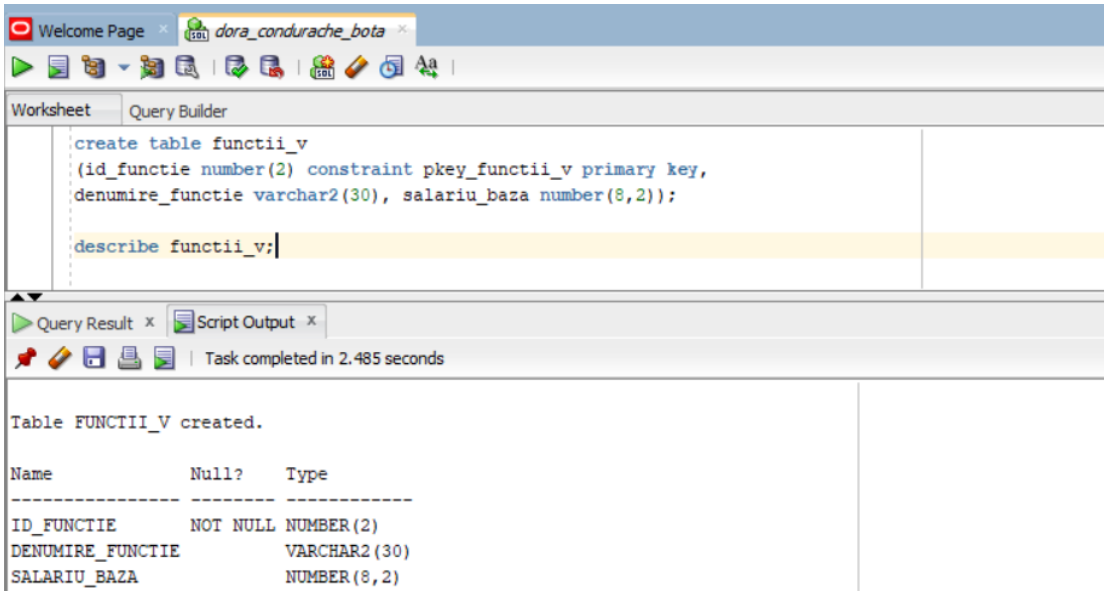


Tabela Clienti_v:

```
CREATE TABLE clienti_v
(id_client number(2) constraint pkey_clienti_v primary key,
prenume varchar2(20), nume varchar2(20),
telefon varchar2(20),email varchar2(30), data_nastere date,
sex varchar(1), preferinte varchar2(40), puncte_fidelitate number(3));
```

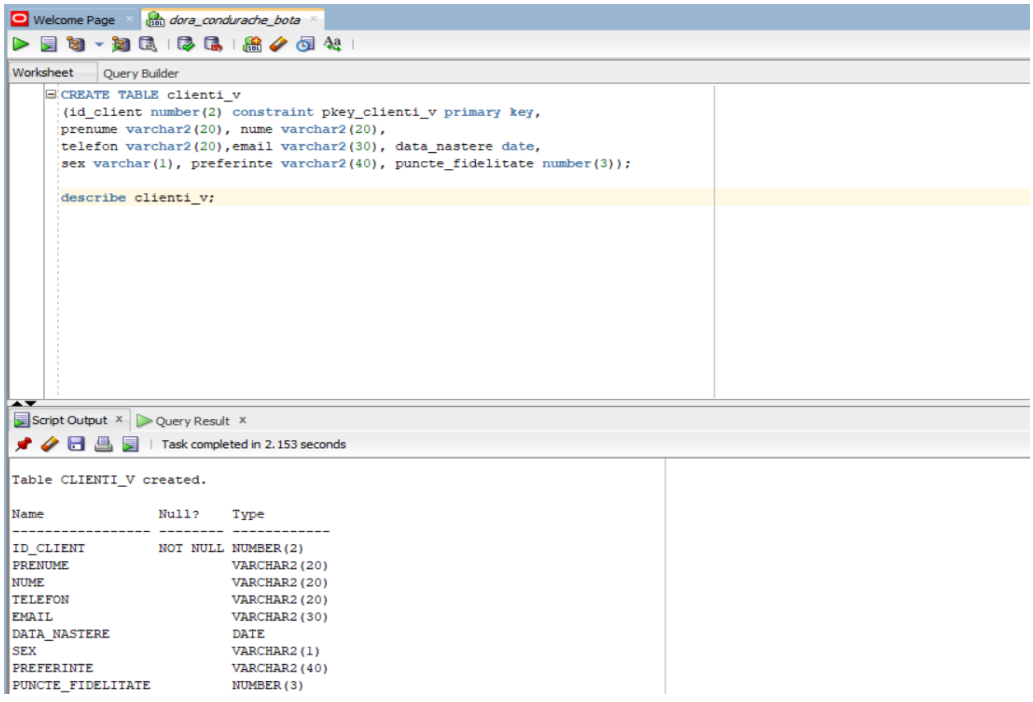
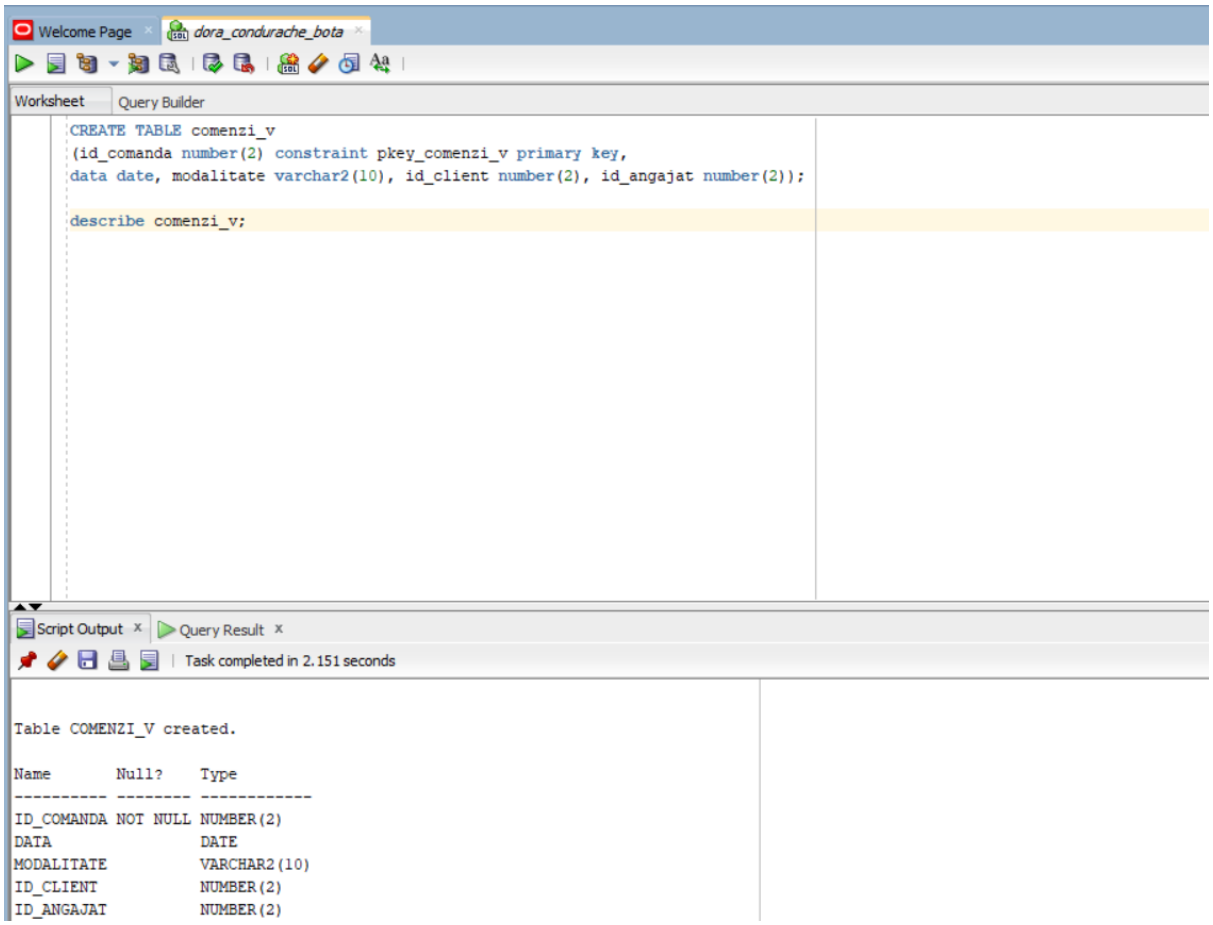


Tabela Comenzi_v:

CREATE TABLE comenzi_v
(id_comanda number(2) constraint pkey_comenzi_v primary key,
data date, modalitate varchar2(10), id_client number(2), id_angajat number(2));

describe comenzi_v;



alter table comenzi_v
add constraint fkey_id_angajat_comenzi_v foreign key (id_angajat)
references angajati_v (id_angajat);

alter table comenzi_v
add constraint fkey_id_client_comenzi_v foreign key (id_client)
references clienti_v (id_client);

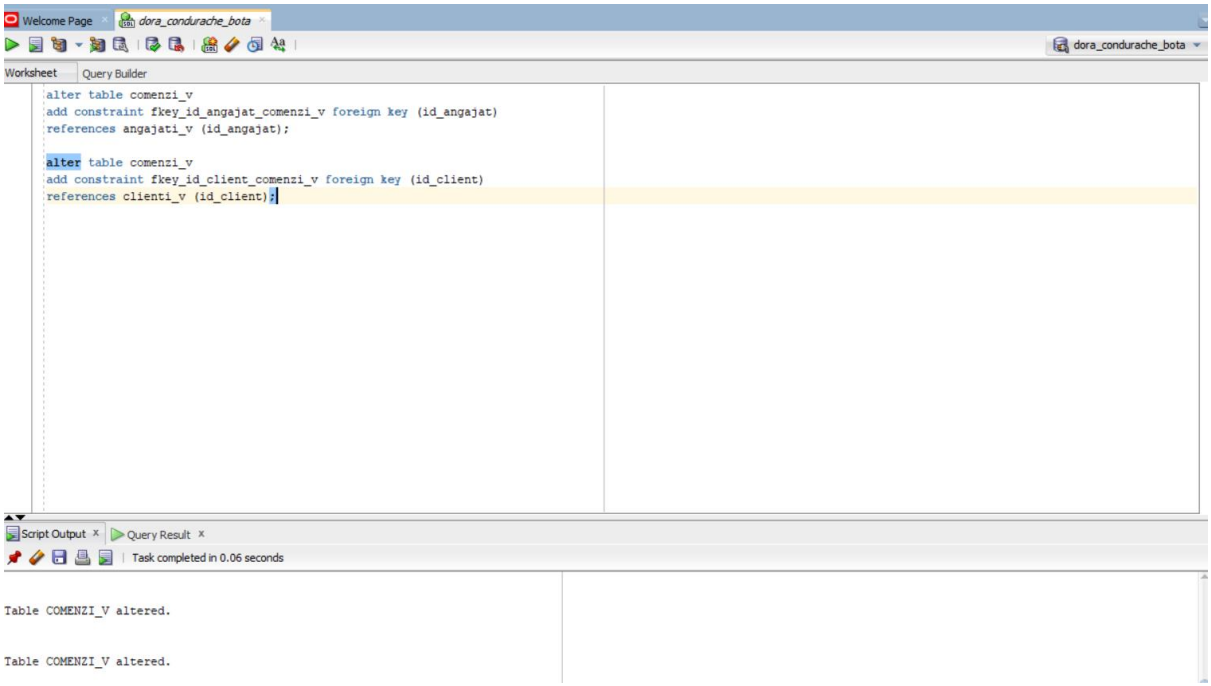


Tabela Produse_v:

CREATE TABLE produse_v
(id_produs number(2) constraint pkey_produse_v primary key,
denumire_produs varchar2(30), descriere varchar2(70),
categorie varchar2(20), personalizabil varchar(2));

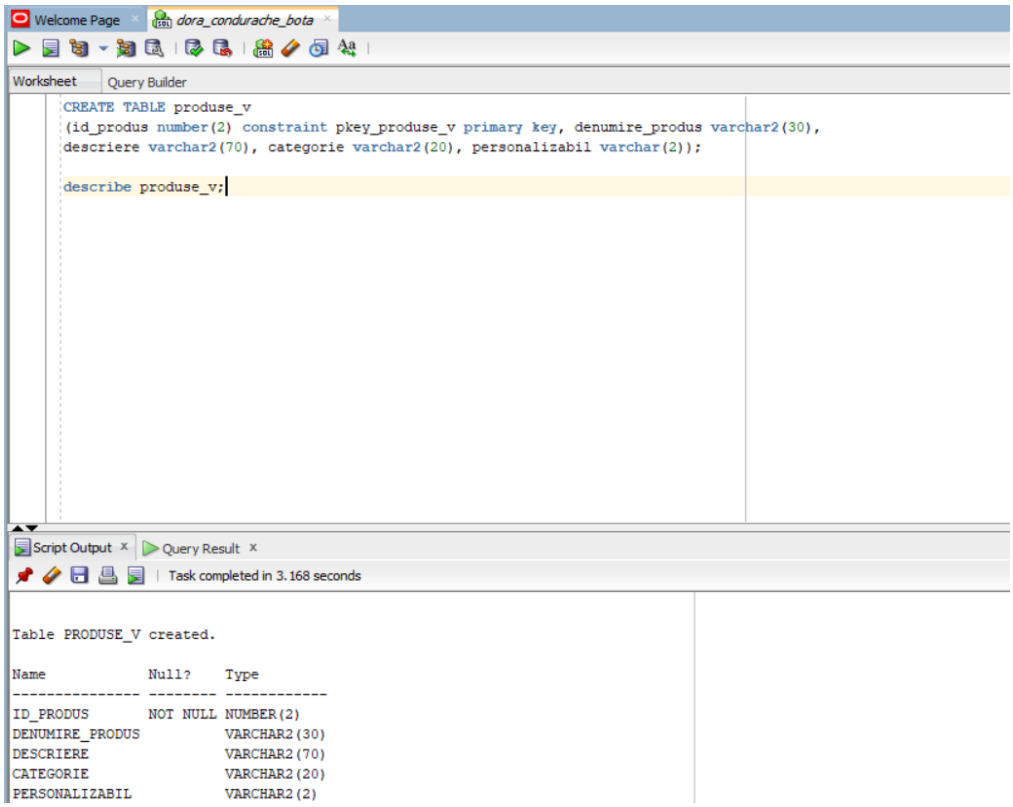
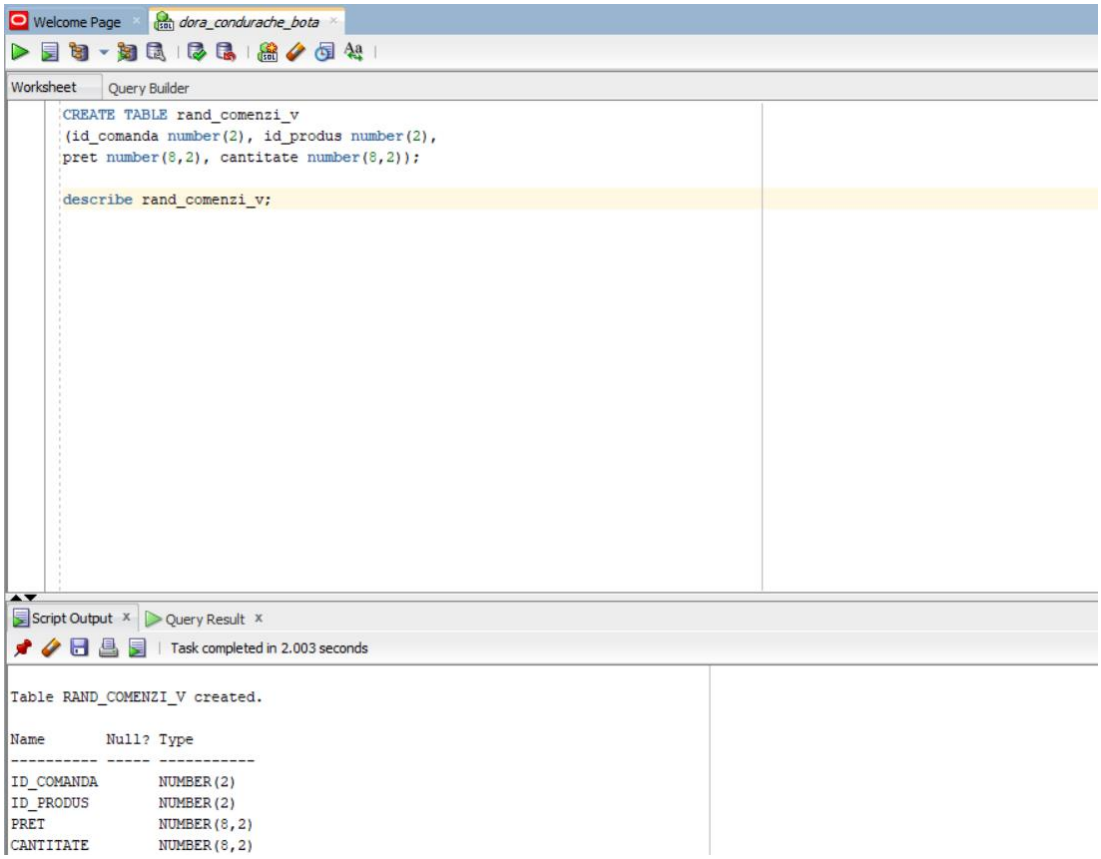


Tabela Rand_comenzi_v:

CREATE TABLE rand_comenzi_v
(id_comanda number(2), id_produs number(2),
pret number(8,2), cantitate number(8,2));



alter table rand_comenzi_v
add constraint fkey_id_comanda_rc_v foreign key (id_comanda)
references comenzi_v (id_comanda);

alter table rand_comenzi_v
add constraint fkey_id_produs_rc_v foreign key (id_produs)
references produse_v (id_produs);

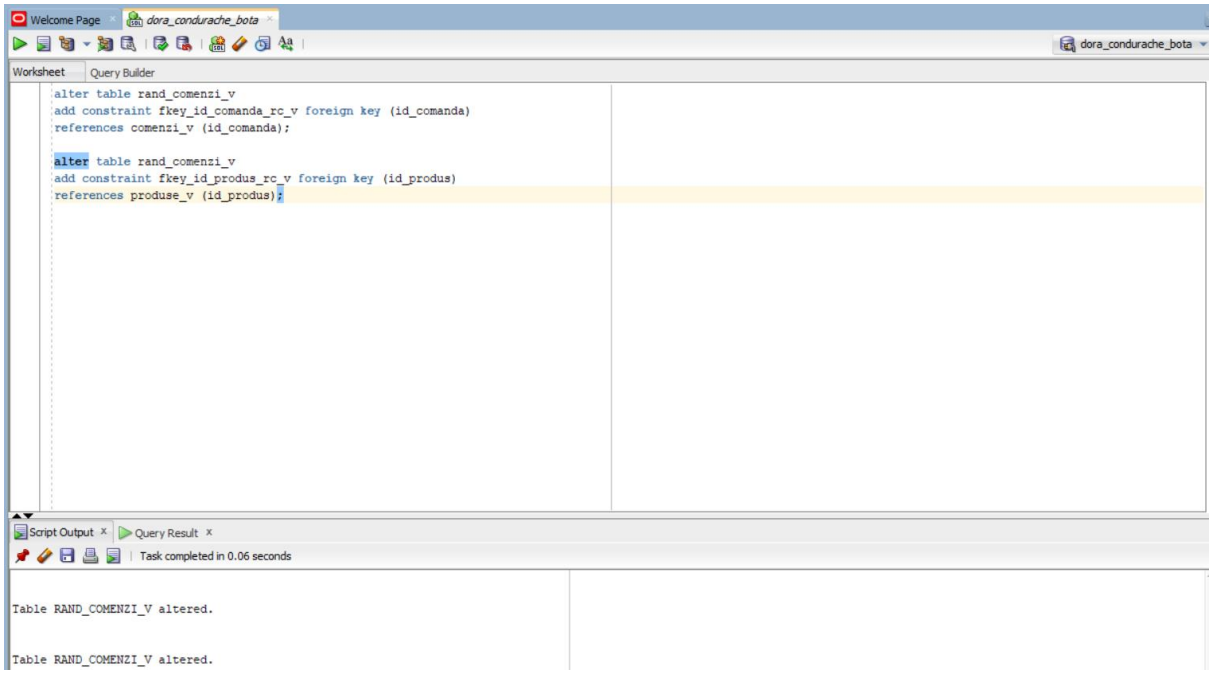


Tabela Retete_v:

CREATE TABLE retete_v
(id_reteta number(2) constraint pkey_retete_v primary key, denumire_reteta varchar2(30), descriere varchar2(60), timp_preparare number(3));

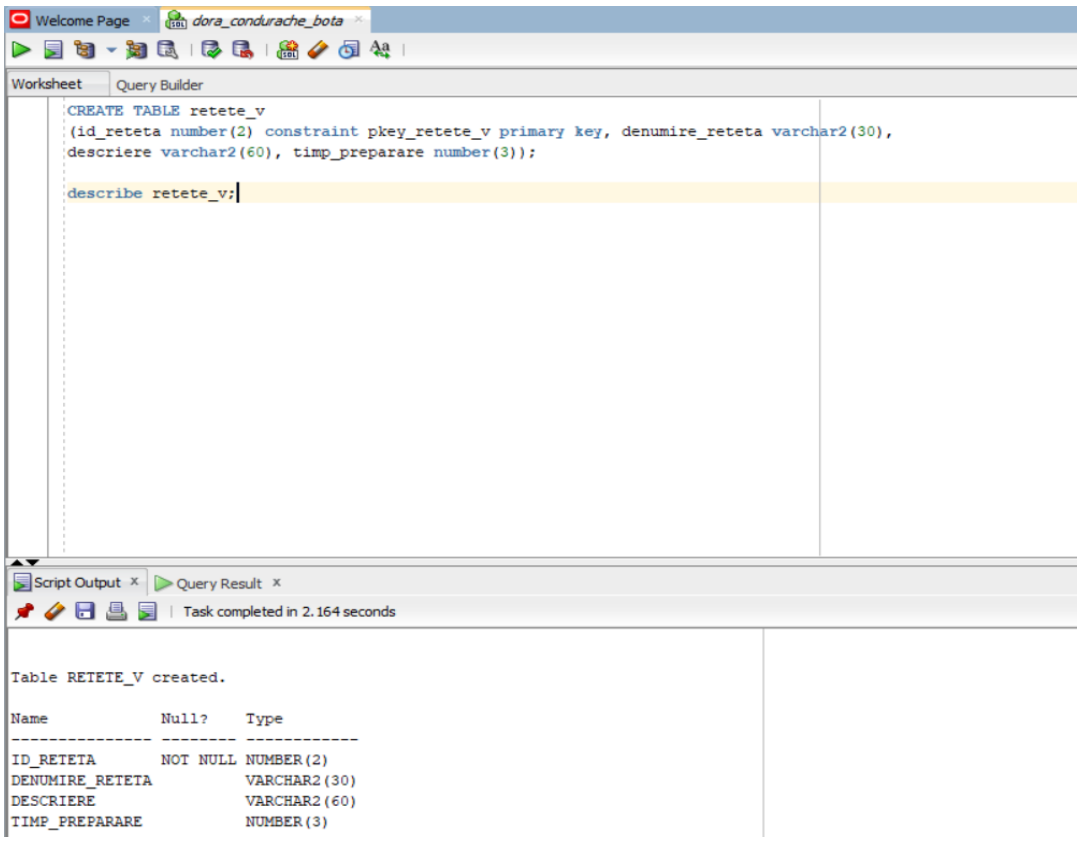


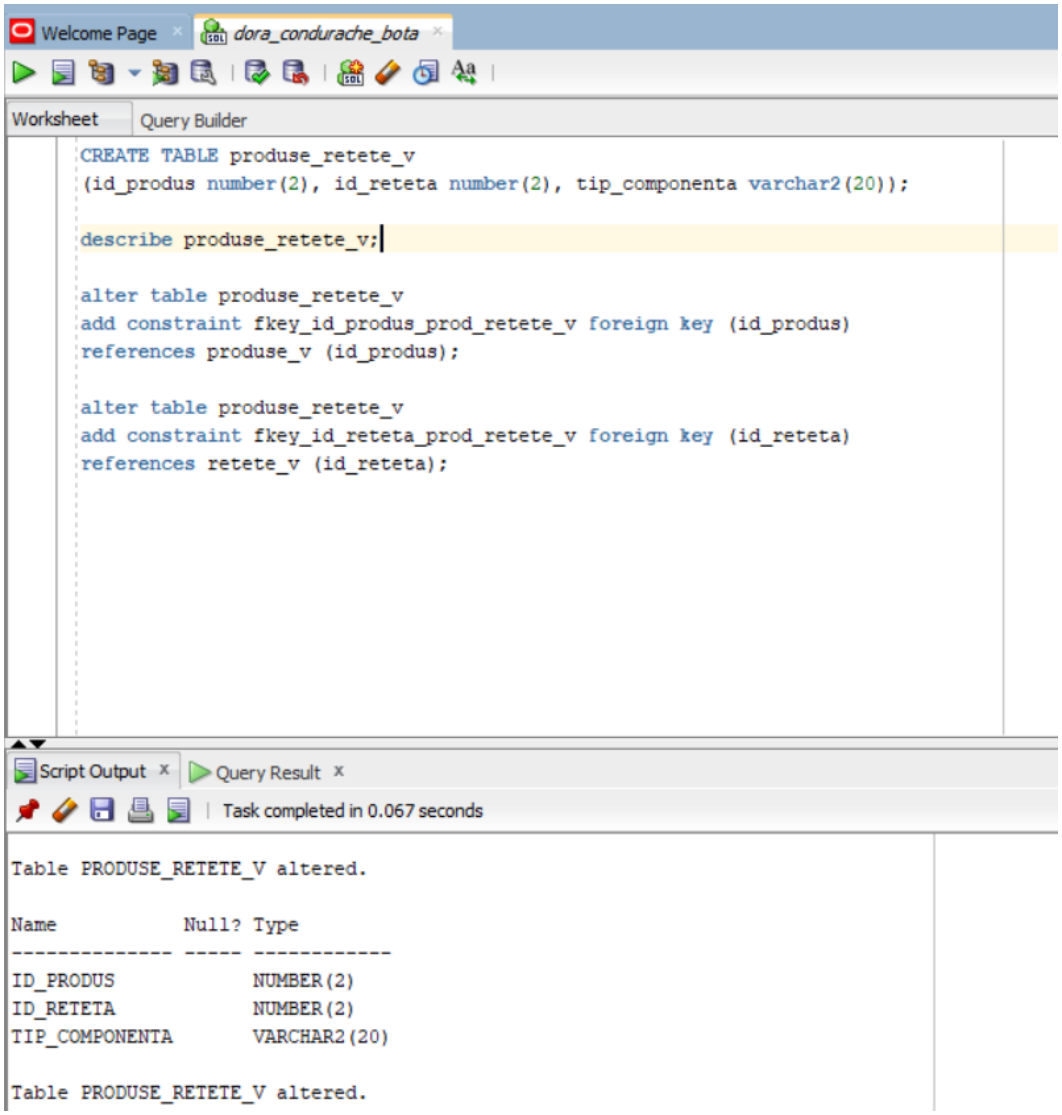
Tabela Produse_retete_v:

CREATE TABLE produse_retete_v
(id_produs number(2), id_reteta number(2), tip_componenta varchar2(20));

describe produse_retete_v;

alter table produse_retete_v
add constraint fkey_id_produs_prod_retete_v foreign key (id_produs)
references produse_v (id_produs);

alter table produse_retete_v
add constraint fkey_id_reteta_prod_retete_v foreign key (id_reteta)
references retete_v (id_reteta);

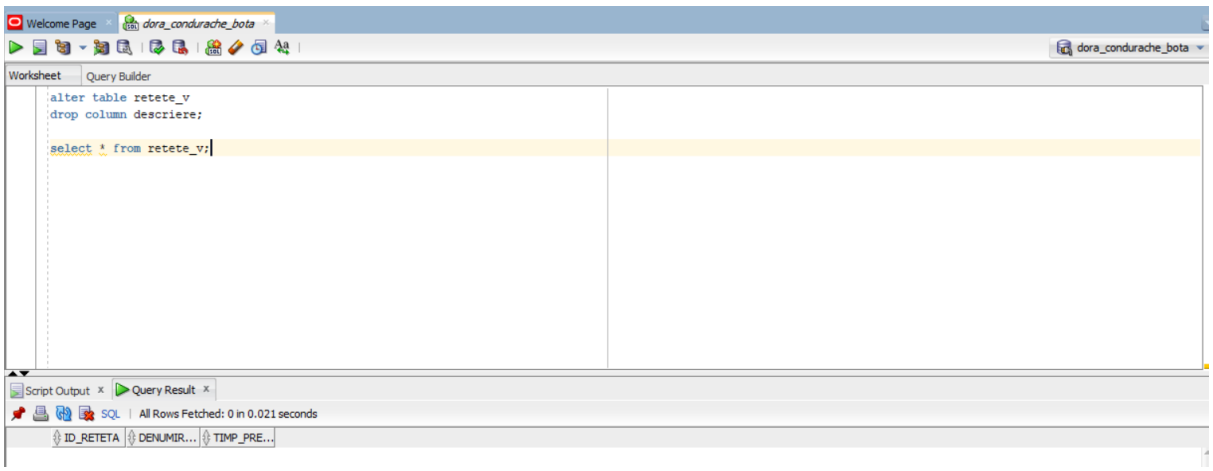


4. Actualizarea structurii tabelelor si modificarea restrictiilor de integritate

1) Eliminăm coloana descriere din tabela Retete_v:

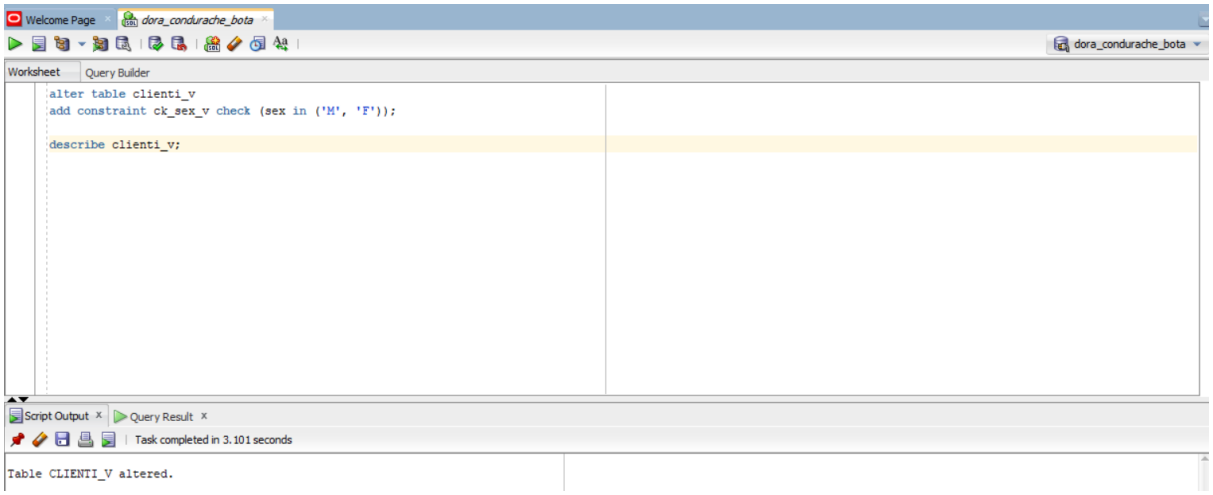
alter table retete_v
drop column descriere;

select * from retete_v;



2) Adăugăm o constrângere CHECK pentru sex în tabela Clienti_v:

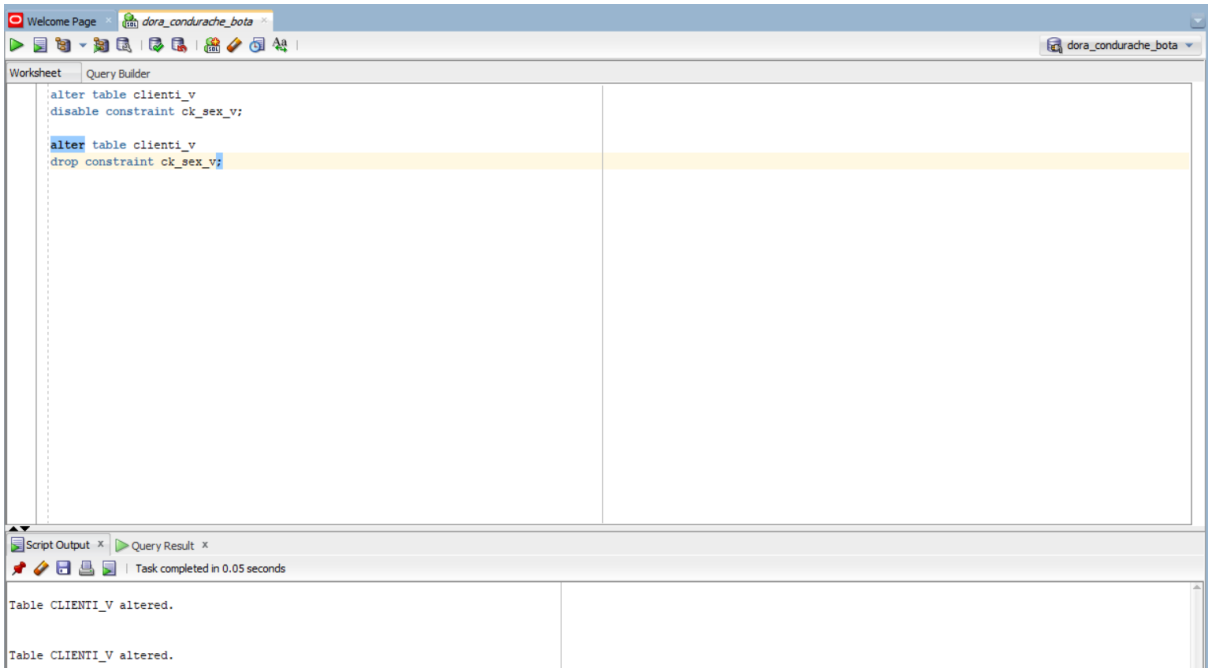
alter table clienti_v
add constraint ck_sex_v check (sex in ('M', 'F'));



3) Dezactivăm și apoi ștergem constrângerea pentru sex în tabela Clienti_v:

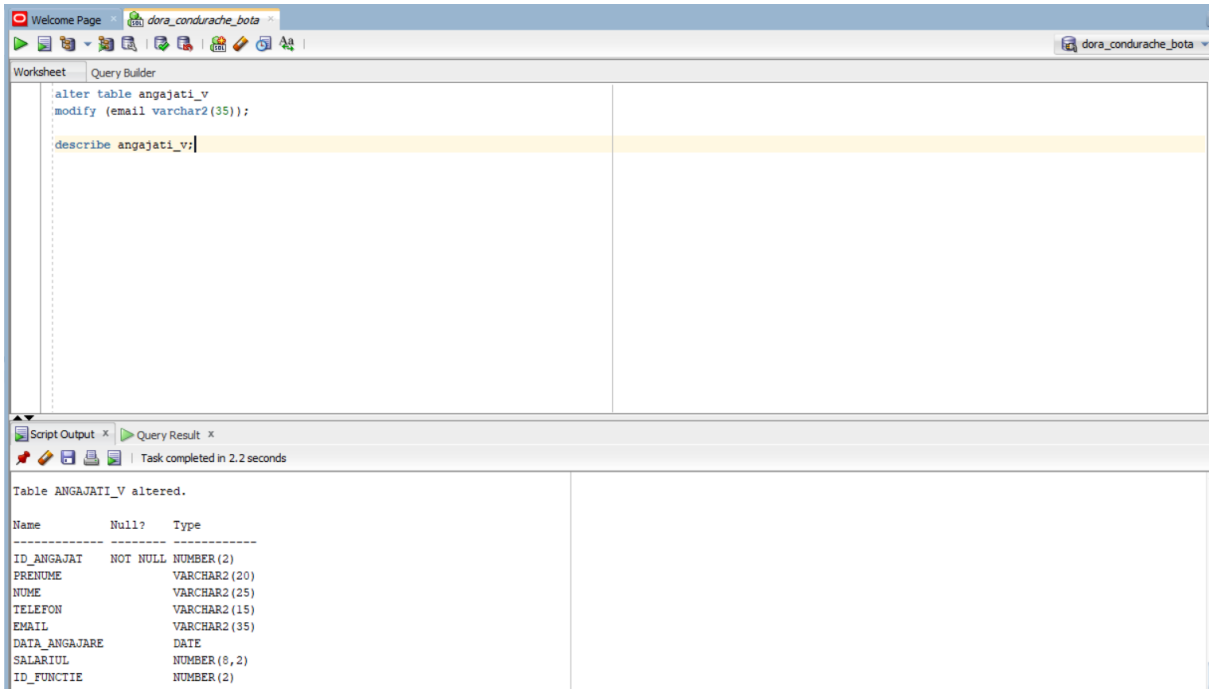
alter table clienti_v
disable constraint ck_sex_v;

alter table clienti_v
drop constraint ck_sex_v;



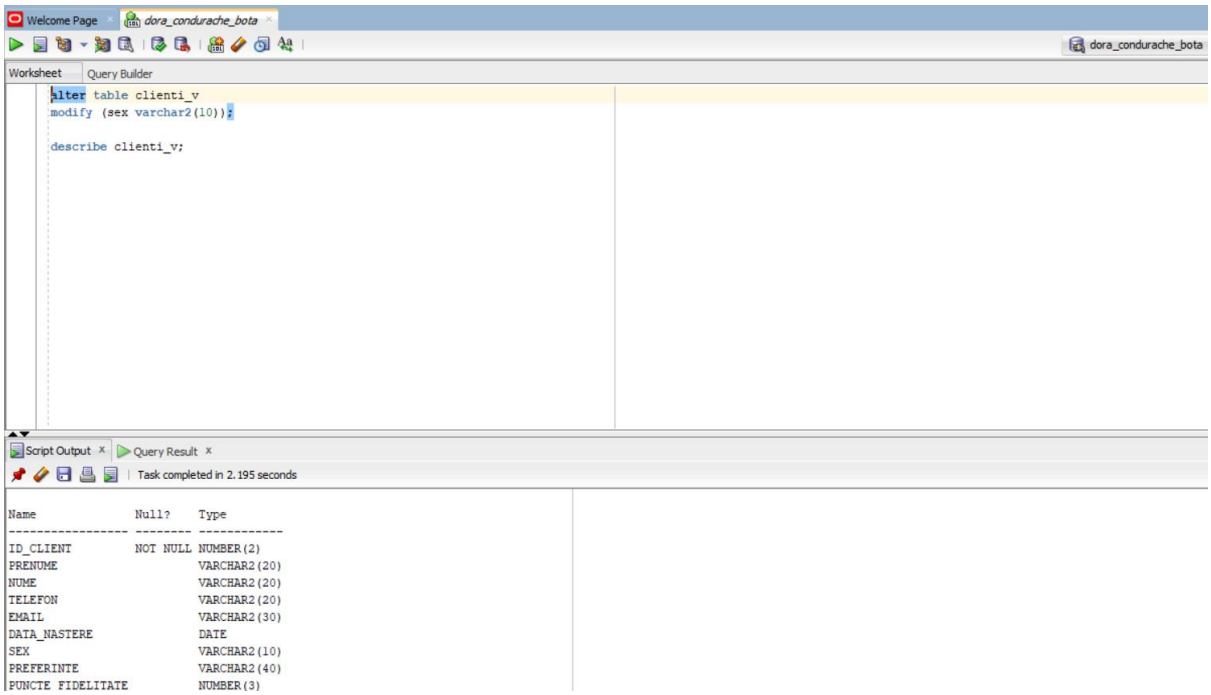
4) Modificăm tipul de date al coloanei email în tabela Angajati_v:

alter table angajati_v
modify (email varchar2(35));



5) Modificăm tipul de date al coloanei sex în tabela Clienti_v:

```
alter table clienti_v
modify (sex varchar2(10));
```



5. Adăugarea (min 10, max 20) de înregistrări în fiecare tabelă

Tabela Angajati_v:

```
INSERT INTO angajati_v VALUES (1, 'Ana', 'Marinescu', '0735264387',
'marinana65@gmail.com', '01-FEB-2019', 5500, 1);
INSERT INTO angajati_v VALUES (2, 'Costel', 'David', '0747231645',
'costeldav4@gmail.com', '24-JUN-2019', 5400, 1);
INSERT INTO angajati_v VALUES (3, 'Crina', 'Popescu', '0754087648',
'crinapop3@yahoo.com', '13-MAR-2023', 4200, 2);
INSERT INTO angajati_v VALUES (4, 'Andra', 'Georgescu', '0734123627',
'anageorg5@gmail.com', '02-JUN-2023', 4300, 2);
INSERT INTO angajati_v VALUES (5, 'Sorina', 'Lungu', '0734175329',
'sorina32@yahoo.com', '08-APR-2023', 4200, 2);
INSERT INTO angajati_v VALUES (6, 'Cristian', 'Vasilescu', '0782598756',
'cristian6@yahoo.com', '19-NOV-2022', 3700, 3);
INSERT INTO angajati_v VALUES (7, 'Codrin', 'Stancu', '0798231593',
'codrin3@gmail.com', '15-OCT-2022', 3800, 3);
INSERT INTO angajati_v VALUES (8, 'Carina', 'Samoila', '0752839654',
'carinasam9@gmail.com', '30-APR-2021', 2700, 4);
```

```
INSERT INTO angajati_v VALUES (9, 'Ioana', 'Marin', '0783175324',
'ioanam8@yahoo.com', '29-APR-2022', 2600, 5);
INSERT INTO angajati_v VALUES (10, 'Roxana', 'Barbu', '0712742956',
'roxanab7@gmail.com', '20-AUG-2021', 3500, 6);
INSERT INTO angajati_v VALUES (11, 'Simona', 'Iliescu', '0728416422',
'simonail2@yahoo.com', '23-SEP-2022', 3600, 6);
INSERT INTO angajati_v VALUES (12, 'Raluca', 'Dobre', '0731768239',
'ralucad8@yahoo.com', '16-SEP-2020', 5300, 7);
INSERT INTO angajati_v VALUES (13, 'Victor', 'Petrescu', '0782317534',
'victorp2@gmail.com', '14-JUN-2021', 5600, 8);
INSERT INTO angajati_v VALUES (14, 'Andreea', 'Popa', '0752142749',
'andreeap5@gmail.com', '21-MAY-2020', 4900, 9);
INSERT INTO angajati_v VALUES (15, 'Corina', 'Muntean', '0728739013',
'corinam6@yahoo.com', '17-JAN-2019', 6000, 10);
INSERT INTO angajati_v VALUES (16, 'Tania', 'Solomon', '0712742361',
'tanias9@gmail.com', '10-SEP-2021', 3600, 6);
INSERT INTO angajati_v VALUES (17, 'Maria', 'Ionascu', '0721749056',
'mariai8@yahoo.com', '06-FEB-2022', 3500, 6);
```

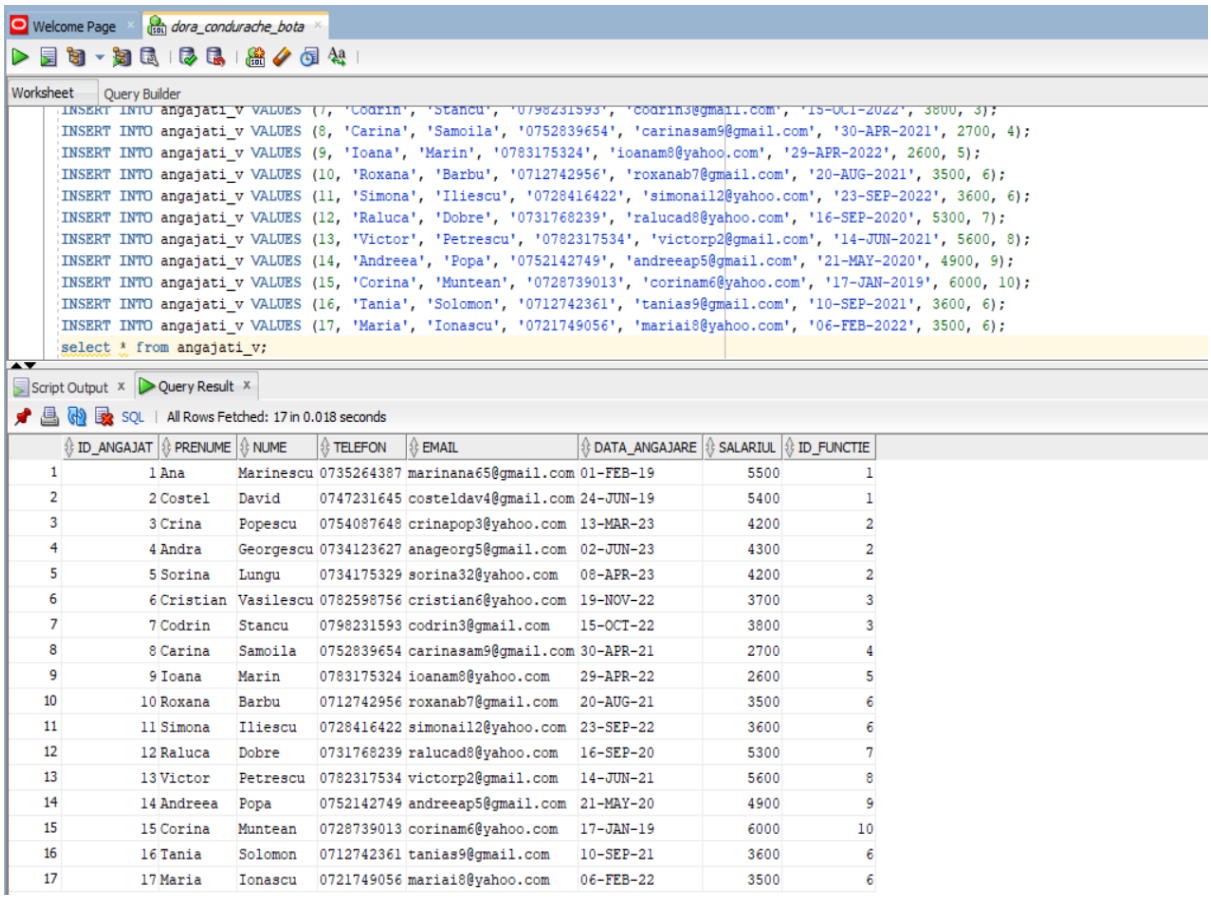
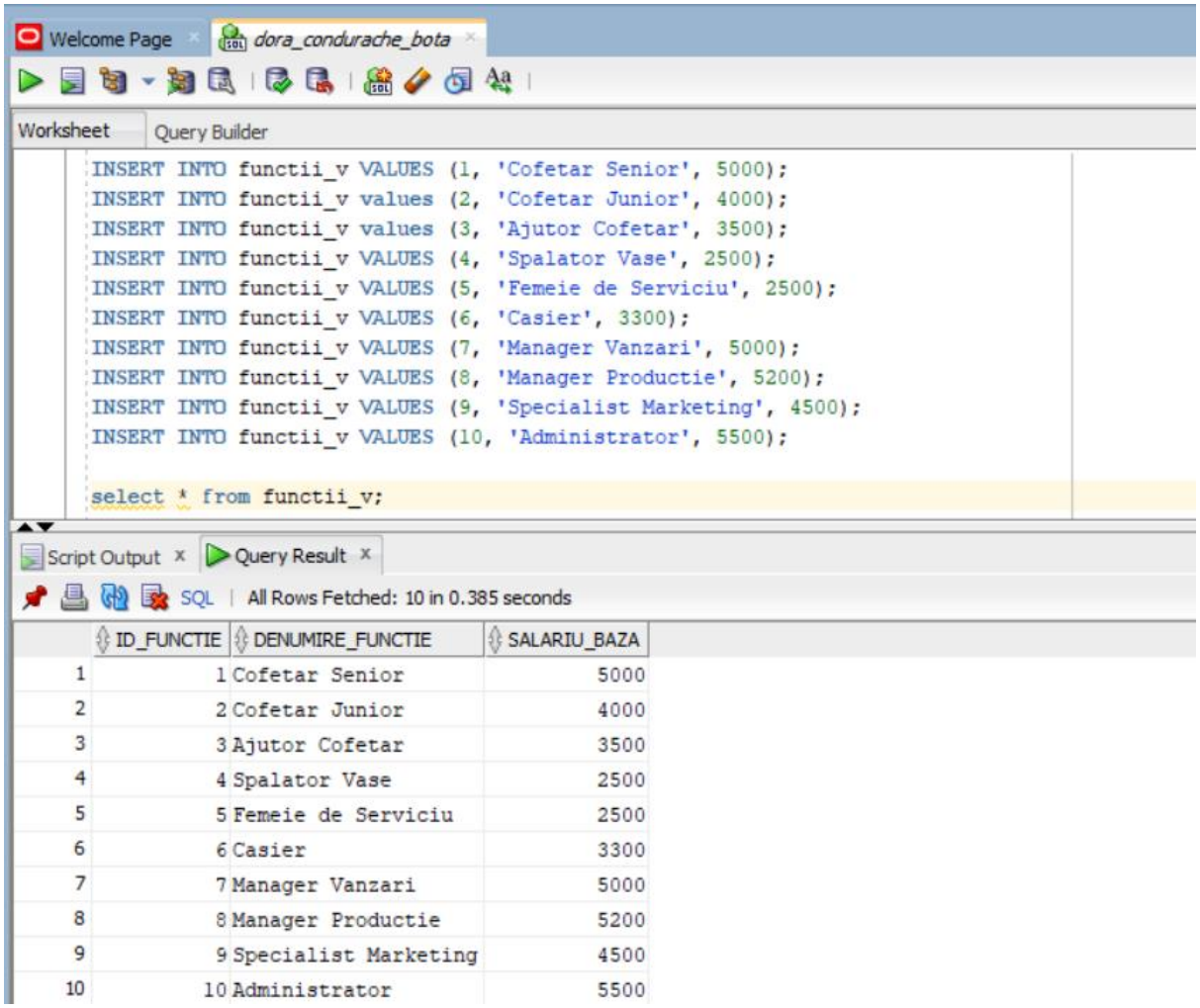


Tabela Functii_v:

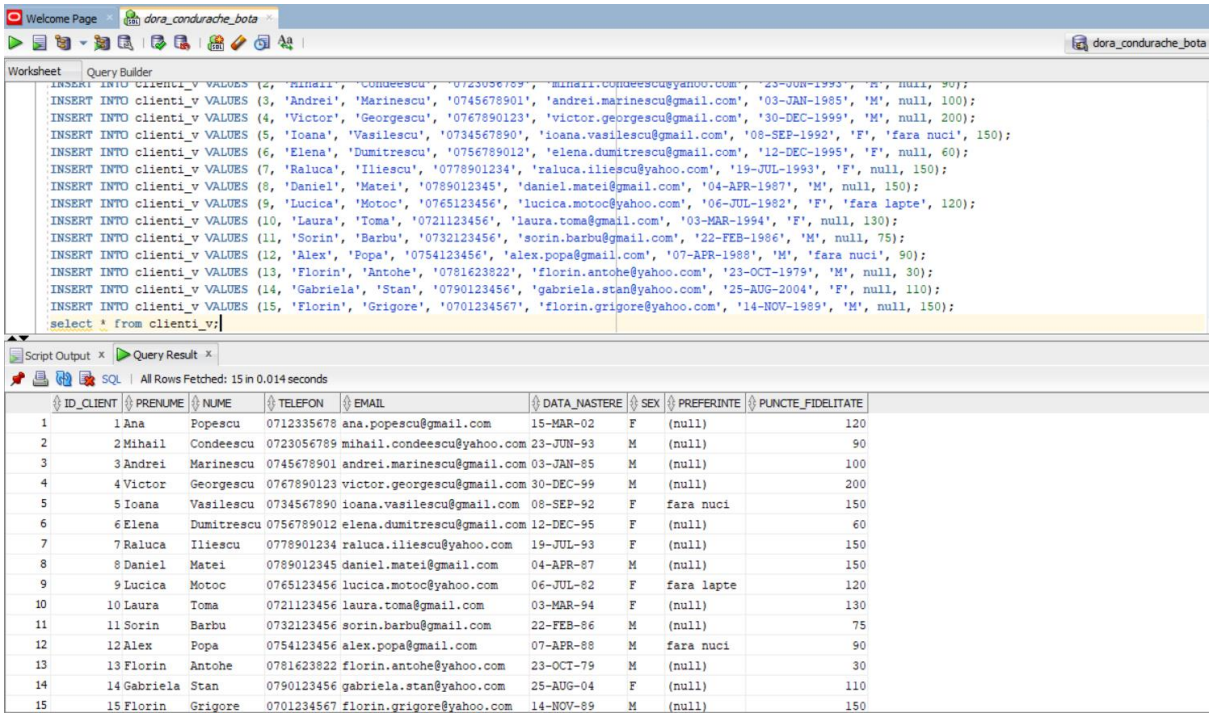
```
INSERT INTO functii_v VALUES (1, 'Cofetar Senior', 5000);
INSERT INTO functii_v VALUES (2, 'Cofetar Junior', 4000);
INSERT INTO functii_v VALUES (3, 'Ajutor Cofetar', 3500);
INSERT INTO functii_v VALUES (4, 'Spalator Vase', 2500);
INSERT INTO functii_v VALUES (5, 'Femeie de Serviciu', 2500);
INSERT INTO functii_v VALUES (6, 'Casier', 3300);
INSERT INTO functii_v VALUES (7, 'Manager Vanzari', 5000);
INSERT INTO functii_v VALUES (8, 'Manager Productie', 5200);
INSERT INTO functii_v VALUES (9, 'Specialist Marketing', 4500);
INSERT INTO functii_v VALUES (10, 'Administrator', 5500);
```



	ID_FUNCTIE	DENUMIRE_FUNCTIE	SALARIU_BAZA
1	1	Cofetar Senior	5000
2	2	Cofetar Junior	4000
3	3	Ajutor Cofetar	3500
4	4	Spalator Vase	2500
5	5	Femeie de Serviciu	2500
6	6	Casier	3300
7	7	Manager Vanzari	5000
8	8	Manager Productie	5200
9	9	Specialist Marketing	4500
10	10	Administrator	5500

Tabela Clienti_v:

```
INSERT INTO clienti_v VALUES (1, 'Ana', 'Popescu', '0712335678',
'ana.popescu@gmail.com', '15-MAR-2002', 'F', null, 120);
INSERT INTO clienti_v VALUES (2, 'Mihail', 'Condeescu', '0723056789',
'mihail.condeescu@yahoo.com', '23-JUN-1993', 'M', null, 90);
INSERT INTO clienti_v VALUES (3, 'Andrei', 'Marinescu', '0745678901',
'andrei.marinescu@gmail.com', '03-JAN-1985', 'M', null, 100);
INSERT INTO clienti_v VALUES (4, 'Victor', 'Georgescu', '0767890123',
'victor.georgescu@gmail.com', '30-DEC-1999', 'M', null, 200);
INSERT INTO clienti_v VALUES (5, 'Ioana', 'Vasilescu', '0734567890',
'ioana.vasilescu@gmail.com', '08-SEP-1992', 'F', 'fara nuci', 150);
INSERT INTO clienti_v VALUES (6, 'Elena', 'Dumitrescu', '0756789012',
'elena.dumitrescu@gmail.com', '12-DEC-1995', 'F', null, 60);
INSERT INTO clienti_v VALUES (7, 'Raluca', 'Iliescu', '0778901234',
'raluca.iliescu@yahoo.com', '19-JUL-1993', 'F', null, 150);
INSERT INTO clienti_v VALUES (8, 'Daniel', 'Matei', '0789012345',
'daniel.matei@gmail.com', '04-APR-1987', 'M', null, 150);
INSERT INTO clienti_v VALUES (9, 'Lucica', 'Motoc', '0765123456',
'lucica.motoc@yahoo.com', '06-JUL-1982', 'F', 'fara lapte', 120);
INSERT INTO clienti_v VALUES (10, 'Laura', 'Toma', '0721123456',
'laura.toma@gmail.com', '03-MAR-1994', 'F', null, 130);
INSERT INTO clienti_v VALUES (11, 'Sorin', 'Barbu', '0732123456',
'sorin.barbu@gmail.com', '22-FEB-1986', 'M', null, 75);
INSERT INTO clienti_v VALUES (12, 'Alex', 'Popa', '0754123456', 'alex.popa@gmail.com',
'07-APR-1988', 'M', 'fara nuci', 90);
INSERT INTO clienti_v VALUES (13, 'Florin', 'Antohe', '0781623822',
'florin.antohe@yahoo.com', '23-OCT-1979', 'M', null, 30);
INSERT INTO clienti_v VALUES (14, 'Gabriela', 'Stan', '0790123456',
'gabriela.stan@yahoo.com', '25-AUG-2004', 'F', null, 110);
INSERT INTO clienti_v VALUES (15, 'Florin', 'Grigore', '0701234567',
'florin.grigore@yahoo.com', '14-NOV-1989', 'M', null, 150);
```



```
INSERT INTO clienti_v VALUES (3, 'Andrei', 'Marinescu', '0745678901', 'andrei.marinescu@gmail.com', '03-JAN-1985', 'M', null, 100);
INSERT INTO clienti_v VALUES (4, 'Victor', 'Georgescu', '0767890123', 'victor.georgescu@gmail.com', '30-DEC-1999', 'M', null, 200);
INSERT INTO clienti_v VALUES (5, 'Ioana', 'Vasilescu', '0734567890', 'ioana.vasilescu@gmail.com', '08-SEP-1992', 'F', 'fara nuci', 150);
INSERT INTO clienti_v VALUES (6, 'Elena', 'Dumitrescu', '0756789012', 'elena.dumitrescu@gmail.com', '12-DEC-1995', 'F', null, 60);
INSERT INTO clienti_v VALUES (7, 'Raluca', 'Iliescu', '0778901234', 'raluca.iliescu@yahoo.com', '19-JUL-1993', 'F', null, 150);
INSERT INTO clienti_v VALUES (8, 'Daniel', 'Matei', '0789012345', 'daniel.matei@gmail.com', '04-APR-1987', 'M', null, 150);
INSERT INTO clienti_v VALUES (9, 'Lucica', 'Motoc', '0765123456', 'lucica.motoc@yahoo.com', '06-JUL-1982', 'F', 'fara lapte', 120);
INSERT INTO clienti_v VALUES (10, 'Laura', 'Toma', '0721123456', 'laura.toma@gmail.com', '03-MAR-1994', 'F', null, 130);
INSERT INTO clienti_v VALUES (11, 'Sorin', 'Barbu', '0732123456', 'sorin.barbu@gmail.com', '22-FEB-1986', 'M', null, 75);
INSERT INTO clienti_v VALUES (12, 'Alex', 'Popa', '0754123456', 'alex.popa@gmail.com', '07-APR-1988', 'M', 'fara nuci', 90);
INSERT INTO clienti_v VALUES (13, 'Florin', 'Antohe', '0781623822', 'florin.antohe@yahoo.com', '23-OCT-1979', 'M', null, 30);
INSERT INTO clienti_v VALUES (14, 'Gabriela', 'Stan', '0790123456', 'gabriela.stan@yahoo.com', '25-AUG-2004', 'F', null, 110);
INSERT INTO clienti_v VALUES (15, 'Florin', 'Grigore', '0701234567', 'florin.grigore@yahoo.com', '14-NOV-1989', 'M', null, 150);
select * from clienti_v;
```

	ID_CLIENT	PRENUME	NUME	TELEFON	EMAIL	DATA_NASTERE	SEX	PREFERINTE	PUNCTE_FIDELITATE
1	1	Ana	Popescu	0712335678	ana.popescu@gmail.com	15-MAR-02	F	(null)	120
2	2	Mihail	Condeescu	0723056789	mihail.condeescu@yahoo.com	23-JUN-93	M	(null)	90
3	3	Andrei	Marinescu	0745678901	andrei.marinescu@gmail.com	03-JAN-85	M	(null)	100
4	4	Victor	Georgescu	0767890123	victor.georgescu@gmail.com	30-DEC-99	M	(null)	200
5	5	Ioana	Vasilescu	0734567890	ioana.vasilescu@gmail.com	08-SEP-92	F	fara nuci	150
6	6	Elena	Dumitrescu	0756789012	elena.dumitrescu@gmail.com	12-DEC-95	F	(null)	60
7	7	Raluca	Iliescu	0778901234	raluca.iliescu@yahoo.com	19-JUL-93	F	(null)	150
8	8	Daniel	Matei	0789012345	daniel.matei@gmail.com	04-APR-87	M	(null)	150
9	9	Lucica	Motoc	0765123456	lucica.motoc@yahoo.com	06-JUL-82	F	fara lapte	120
10	10	Laura	Toma	0721123456	laura.toma@gmail.com	03-MAR-94	F	(null)	130
11	11	Sorin	Barbu	0732123456	sorin.barbu@gmail.com	22-FEB-86	M	(null)	75
12	12	Alex	Popa	0754123456	alex.popa@gmail.com	07-APR-88	M	fara nuci	90
13	13	Florin	Antohe	0781623822	florin.antohe@yahoo.com	23-OCT-79	M	(null)	30
14	14	Gabriela	Stan	0790123456	gabriela.stan@yahoo.com	25-AUG-04	F	(null)	110
15	15	Florin	Grigore	0701234567	florin.grigore@yahoo.com	14-NOV-89	M	(null)	150

Tabela Comenzi_v:

INSERT INTO comenzi_v VALUES (1, TO_DATE('03-NOV-2024', 'DD-MON-YYYY'), 'fizic', 1, 10);
INSERT INTO comenzi_v VALUES (2, TO_DATE('05-NOV-2024', 'DD-MON-YYYY'), 'online', 2, 17);
INSERT INTO comenzi_v VALUES (3, TO_DATE('08-NOV-2024', 'DD-MON-YYYY'), 'fizic', 3, 11);
INSERT INTO comenzi_v VALUES (4, TO_DATE('10-NOV-2024', 'DD-MON-YYYY'), 'fizic', 3, 16);
INSERT INTO comenzi_v VALUES (5, TO_DATE('12-NOV-2024', 'DD-MON-YYYY'), 'fizic', 8, 10);
INSERT INTO comenzi_v VALUES (6, TO_DATE('14-NOV-2024', 'DD-MON-YYYY'), 'online', 6, 10);
INSERT INTO comenzi_v VALUES (7, TO_DATE('16-NOV-2024', 'DD-MON-YYYY'), 'fizic', 7, 11);
INSERT INTO comenzi_v VALUES (8, TO_DATE('18-NOV-2024', 'DD-MON-YYYY'), 'online', 8, 16);
INSERT INTO comenzi_v VALUES (9, TO_DATE('20-NOV-2024', 'DD-MON-YYYY'), 'fizic', 11, 17);
INSERT INTO comenzi_v VALUES (10, TO_DATE('22-NOV-2024', 'DD-MON-YYYY'), 'fizic', 10, 11);
INSERT INTO comenzi_v VALUES (11, TO_DATE('23-NOV-2024', 'DD-MON-YYYY'), 'fizic', 15, 17);

INSERT INTO comenzi_v VALUES (12, TO_DATE('28-NOV-2024', 'DD-MON-YYYY'), 'online', 12, 16);
INSERT INTO comenzi_v VALUES (13, TO_DATE('30-NOV-2024', 'DD-MON-YYYY'), 'fizic', 14, 17);
INSERT INTO comenzi_v VALUES (14, TO_DATE('02-DEC-2024', 'DD-MON-YYYY'), 'online', 2, 10);
INSERT INTO comenzi_v VALUES (15, TO_DATE('03-DEC-2024', 'DD-MON-YYYY'), 'fizic', 15, 17);

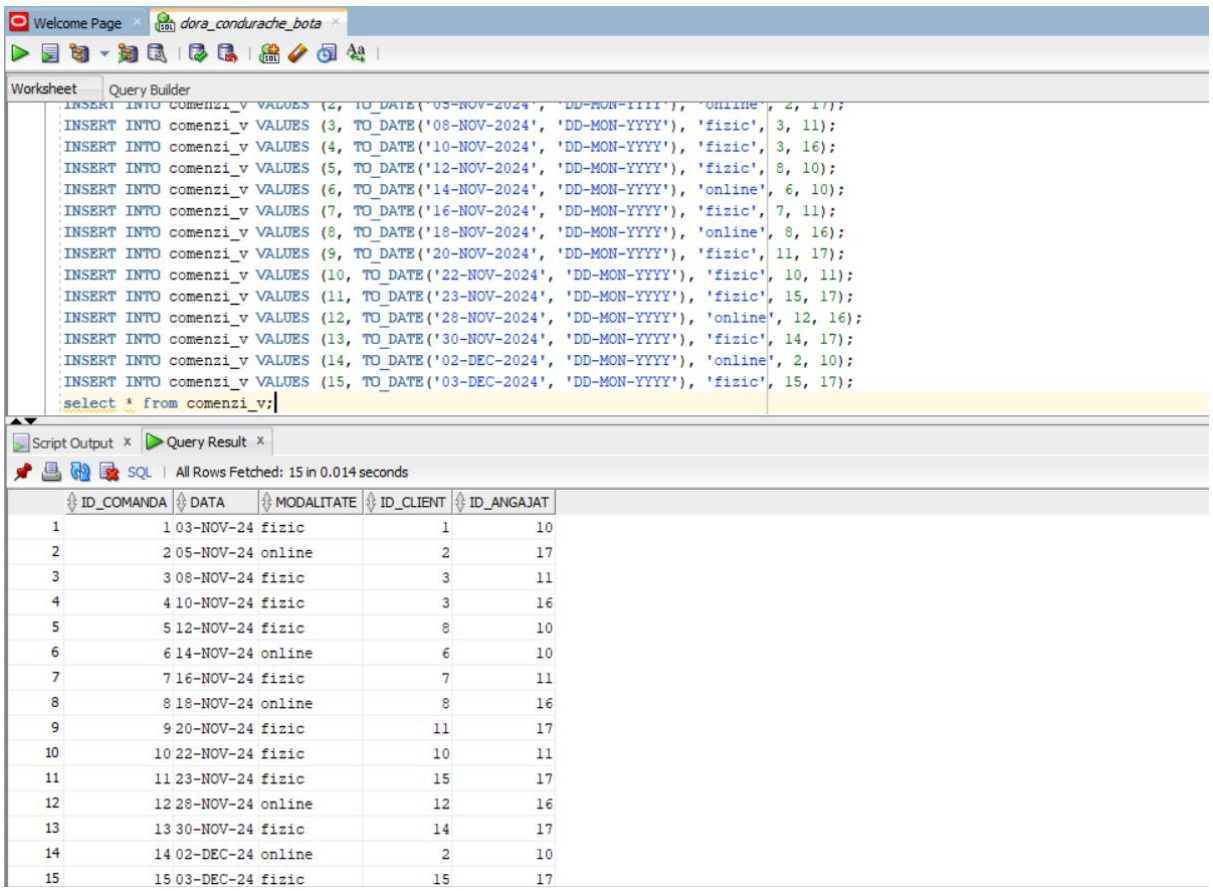


Tabela Produse_v:

INSERT INTO produse_v VALUES (1, 'Tort aniversar', 'Tort personalizabil pentru aniversari', 'Torturi', 'DA');
INSERT INTO produse_v VALUES (2, 'Tort egiptean', 'Tort cu blat cu nuci si crema de caramel', 'Torturi', 'NU');
INSERT INTO produse_v VALUES (3, 'Tort cu fructe', 'Tort cu blat de vanilie si fructe proaspete', 'Torturi', 'DA');
INSERT INTO produse_v VALUES (4, 'Tiramisu', 'Desert italian cu mascarpone si cafea', 'Prajituri', 'NU');
INSERT INTO produse_v VALUES (5, 'Cheesecake capsuni', 'Cheesecake copt cu topping de capsuni', 'Torturi', 'DA');

INSERT INTO produse_v VALUES (6, 'Eclere', 'Eclere cu crema de vanilie si glazura de ciocolata', 'Prajituri', 'NU');
INSERT INTO produse_v VALUES (7, 'Tort ciocolata', 'Tort decadent cu crema de ciocolata belgiana', 'Torturi', 'DA');
INSERT INTO produse_v VALUES (8, 'Red Velvet', 'Tort cu blat rosu catifelat si crema de branza', 'Torturi', 'NU');
INSERT INTO produse_v VALUES (9, 'Tort Krantz', 'Tort cu crema de unt si caramel crocant', 'Torturi', 'NU');
INSERT INTO produse_v VALUES (10, 'Briose', 'Briose pufoase cu diverse arome', 'Prajituri', 'DA');
INSERT INTO produse_v VALUES (11, 'Minitarte cu fructe', 'Tarte mici cu crema de vanilie si fructe proaspete', 'Prajituri', 'DA');
INSERT INTO produse_v VALUES (12, 'Fursecuri', 'Fursecuri crocante cu fulgi de ciocolata', 'Prajituri', 'DA');
INSERT INTO produse_v VALUES (13, 'Macarons', 'Macarons colorate si aromate, umplute cu crema', 'Prajituri', 'DA');
INSERT INTO produse_v VALUES (14, 'Cannoli', 'Dulciuri italiene umplute cu crema', 'Prajituri', 'NU');
INSERT INTO produse_v VALUES (15, 'Cake Pops', 'Acadele colorate din blat de ciocolata', 'Prajituri', 'DA');

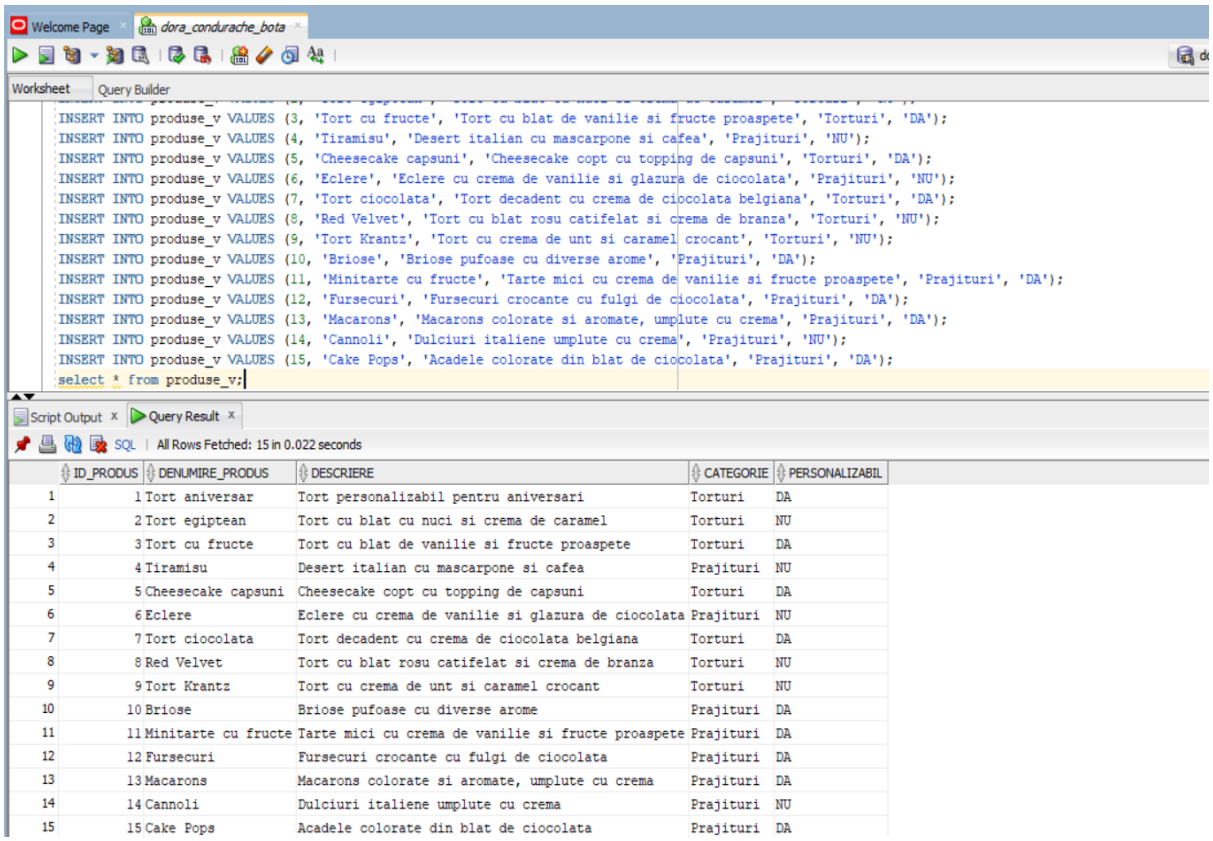
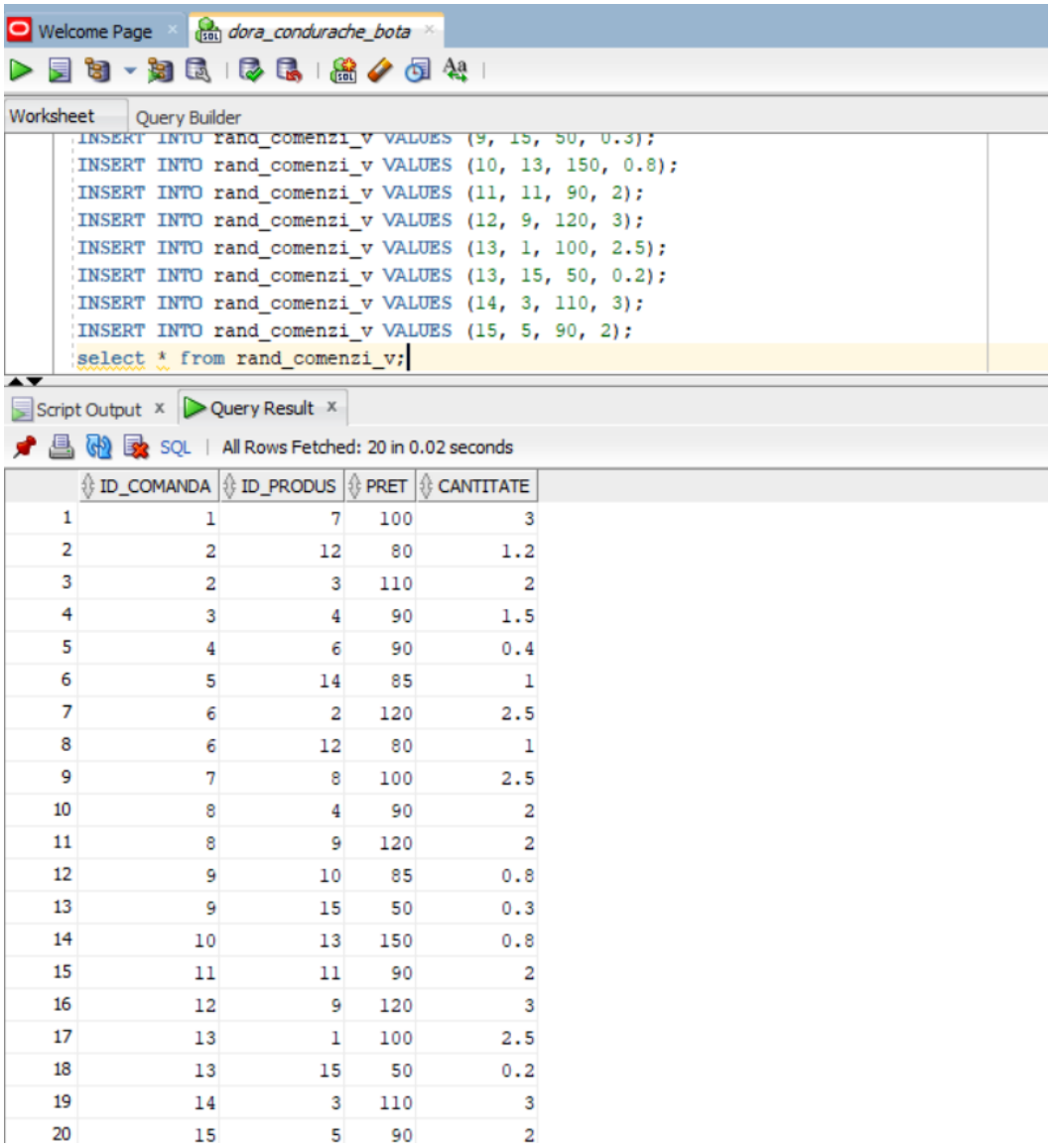


Tabela Rand_comenzi_v:

```
INSERT INTO rand_comenzi_v VALUES (1, 7, 100, 3);
INSERT INTO rand_comenzi_v VALUES (2, 12, 80, 1.2);
INSERT INTO rand_comenzi_v VALUES (2, 3, 110, 2);
INSERT INTO rand_comenzi_v VALUES (3, 4, 90, 1.5);
INSERT INTO rand_comenzi_v VALUES (4, 6, 90, 0.4);
INSERT INTO rand_comenzi_v VALUES (5, 14, 85, 1);
INSERT INTO rand_comenzi_v VALUES (6, 2, 120, 2.5);
INSERT INTO rand_comenzi_v VALUES (6, 12, 80, 1);
INSERT INTO rand_comenzi_v VALUES (7, 8, 100, 2.5);
INSERT INTO rand_comenzi_v VALUES (8, 4, 90, 2);
INSERT INTO rand_comenzi_v VALUES (8, 9, 120, 2);
INSERT INTO rand_comenzi_v VALUES (9, 10, 85, 0.8);
INSERT INTO rand_comenzi_v VALUES (9, 15, 50, 0.3);
INSERT INTO rand_comenzi_v VALUES (10, 13, 150, 0.8);
INSERT INTO rand_comenzi_v VALUES (11, 11, 90, 2);
INSERT INTO rand_comenzi_v VALUES (12, 9, 120, 3);
INSERT INTO rand_comenzi_v VALUES (13, 1, 100, 2.5);
INSERT INTO rand_comenzi_v VALUES (13, 15, 50, 0.2);
INSERT INTO rand_comenzi_v VALUES (14, 3, 110, 3);
INSERT INTO rand_comenzi_v VALUES (15, 5, 90, 2);
```



	ID_COMANDA	ID_PRODUS	PRET	CANTITATE
1	1	7	100	3
2	2	12	80	1.2
3	2	3	110	2
4	3	4	90	1.5
5	4	6	90	0.4
6	5	14	85	1
7	6	2	120	2.5
8	6	12	80	1
9	7	8	100	2.5
10	8	4	90	2
11	8	9	120	2
12	9	10	85	0.8
13	9	15	50	0.3
14	10	13	150	0.8
15	11	11	90	2
16	12	9	120	3
17	13	1	100	2.5
18	13	15	50	0.2
19	14	3	110	3
20	15	5	90	2

Tabela Retete_v:

INSERT INTO retete_v VALUES (1, 'Blat de vanilie', 45);
INSERT INTO retete_v VALUES (2, 'Blat de ciocolata', 50);
INSERT INTO retete_v VALUES (3, 'Blat Red Velvet', 50);
INSERT INTO retete_v VALUES (4, 'Blat cu nuci', 60);
INSERT INTO retete_v VALUES (5, 'Crema de vanilie', 25);
INSERT INTO retete_v VALUES (6, 'Crema de ciocolata', 30);
INSERT INTO retete_v VALUES (7, 'Crema mascarpone', 40);
INSERT INTO retete_v VALUES (8, 'Crema cu fructe', 45);
INSERT INTO retete_v VALUES (9, 'Macarons', 120);
INSERT INTO retete_v VALUES (10, 'Cake Pops', 70);
INSERT INTO retete_v VALUES (11, 'Briose', 80);
INSERT INTO retete_v VALUES (12, 'Crema de caramel', 50);

INSERT INTO retete_v VALUES (13, 'Fursecuri', 50);
INSERT INTO retete_v VALUES (14, 'Blat tarte', 30);

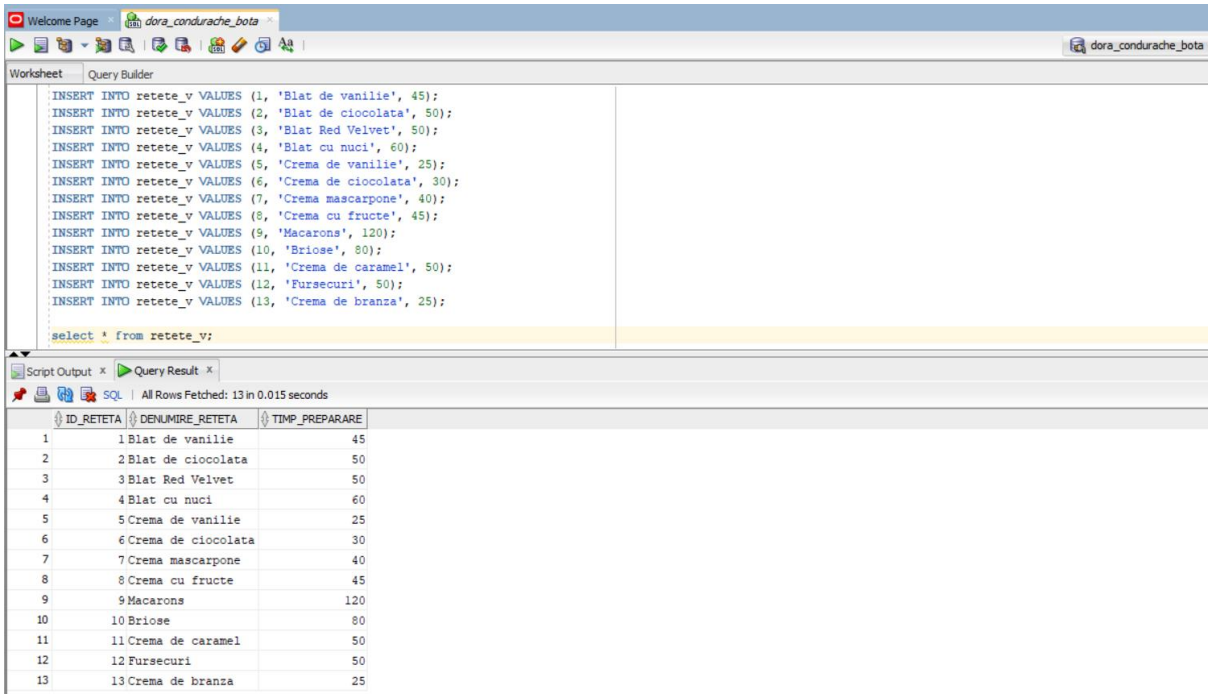
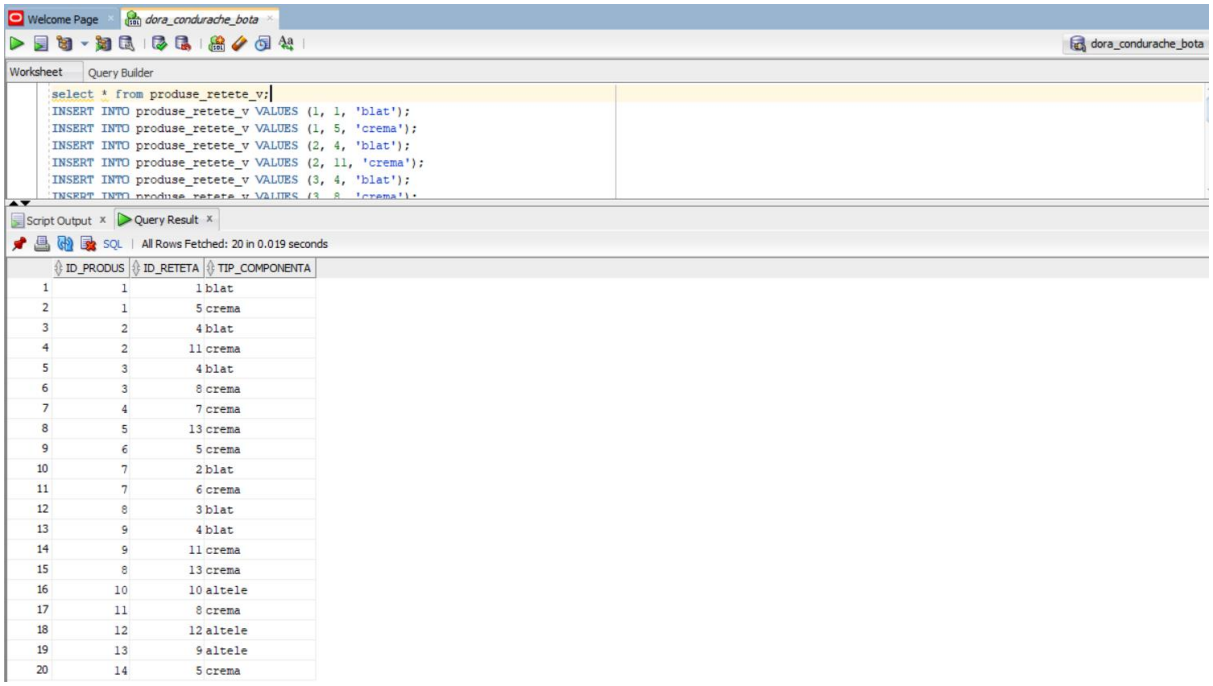


Tabela Produse_retete_v:

INSERT INTO produse_retete_v VALUES (1, 1, 'blat');
INSERT INTO produse_retete_v VALUES (1, 5, 'crema');
INSERT INTO produse_retete_v VALUES (2, 4, 'blat');
INSERT INTO produse_retete_v VALUES (2, 11, 'crema');
INSERT INTO produse_retete_v VALUES (3, 4, 'blat');
INSERT INTO produse_retete_v VALUES (3, 8, 'crema');
INSERT INTO produse_retete_v VALUES (4, 7, 'crema');
INSERT INTO produse_retete_v VALUES (5, 13, 'crema');
INSERT INTO produse_retete_v VALUES (6, 5, 'crema');
INSERT INTO produse_retete_v VALUES (7, 2, 'blat');
INSERT INTO produse_retete_v VALUES (7, 6, 'crema');
INSERT INTO produse_retete_v VALUES (8, 3, 'blat');
INSERT INTO produse_retete_v VALUES (8, 14, 'crema');
INSERT INTO produse_retete_v VALUES (9, 4, 'blat');
INSERT INTO produse_retete_v VALUES (9, 11, 'crema');
INSERT INTO produse_retete_v VALUES (10, 10, 'altele');
INSERT INTO produse_retete_v VALUES (11, 8, 'crema');
INSERT INTO produse_retete_v VALUES (12, 12, 'altele');
INSERT INTO produse_retete_v VALUES (13, 9, 'altele');
INSERT INTO produse_retete_v VALUES (14, 5, 'crema');

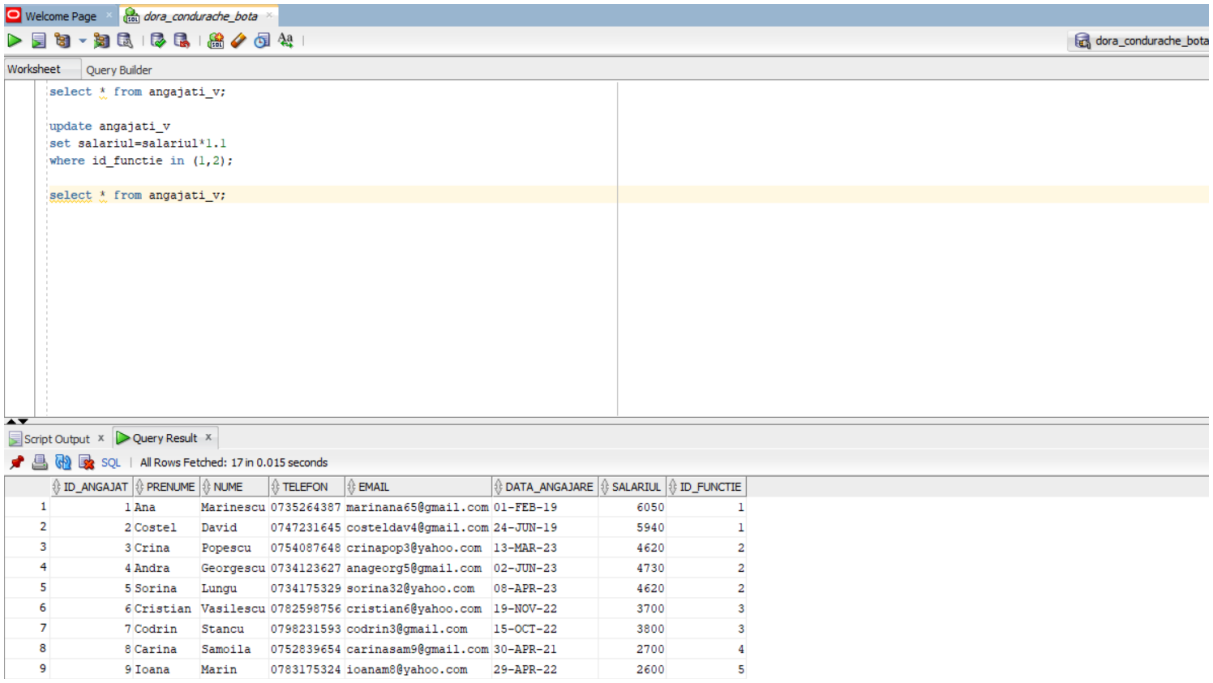


	ID_PRODUS	ID_RETETA	TIP_COMPONENTA
1	1	1	blat
2	1	5	crema
3	2	4	blat
4	2	11	crema
5	3	4	blat
6	3	8	crema
7	4	7	crema
8	5	13	crema
9	6	5	crema
10	7	2	blat
11	7	6	crema
12	8	3	blat
13	9	4	blat
14	9	11	crema
15	8	13	crema
16	10	10	altele
17	11	8	crema
18	12	12	altele
19	13	9	altele
20	14	5	crema

6. Actualizarea înregistrărilor

1) Creștem salariul cofetarilor cu 10%.

```
update angajati_v
set salariul=salariul*1.1
where id_functie in (1,2);
```

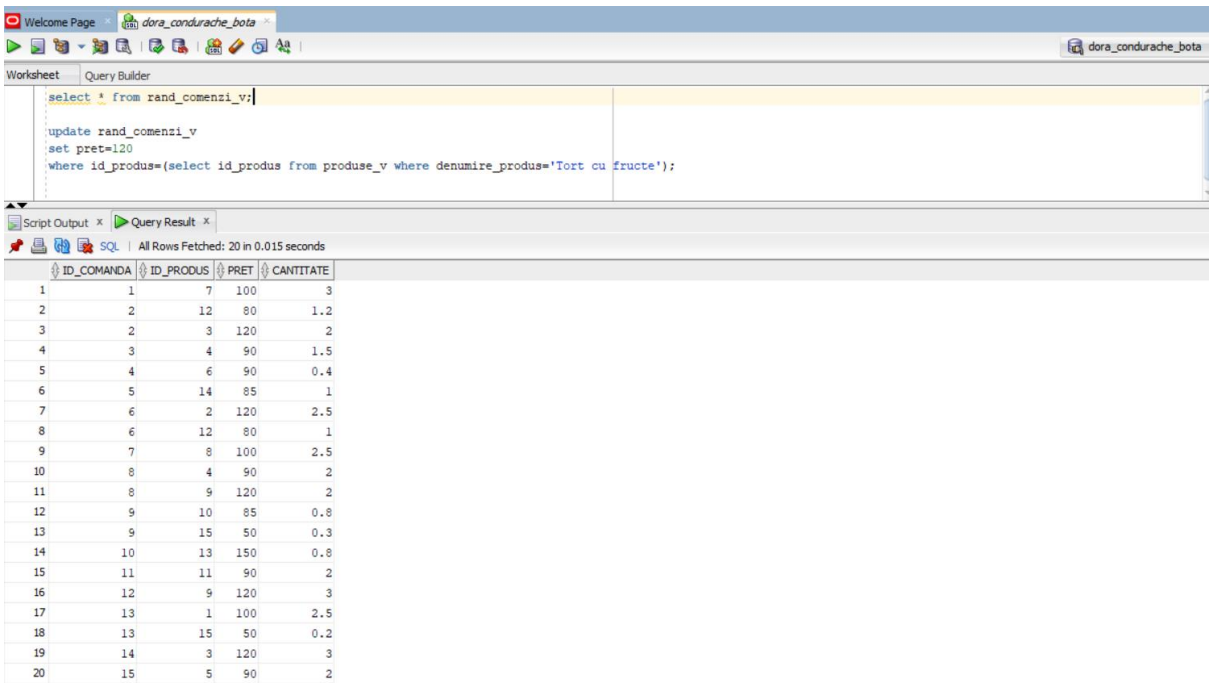


	ID_ANGAJAT	PRENUME	NUME	TELEFON	EMAIL	DATA_ANGAJARE	SALARIUL	ID_FUNCTIE
1	1	Ana	Marinescu	0735264387	marinana65@gmail.com	01-FEB-19	6050	1
2	2	Costel	David	0747231645	costeldav4@gmail.com	24-JUN-19	5940	1
3	3	Crina	Popescu	0754087648	crinapop3@yahoo.com	13-MAR-23	4620	2
4	4	Andra	Georgescu	0734123627	anageorg5@gmail.com	02-JUN-23	4730	2
5	5	Sorina	Lungu	0734175329	sorina32@yahoo.com	08-APR-23	4620	2
6	6	Cristian	Vasilescu	0782598756	cristian6@yahoo.com	19-NOV-22	3700	3
7	7	Codrin	Stancu	0798231593	codrin3@gmail.com	15-OCT-22	3800	3
8	8	Carina	Samoila	0752839654	carinasam9@gmail.com	30-APR-21	2700	4
9	9	Ioana	Marin	0783175324	ioanam8@yahoo.com	29-APR-22	2600	5

2) Corectăm prețul tortului cu fructe de la 110 la 120 (se vor modifica 2 rânduri).

```
update rand_comenzi_v
set pret=120
where id_produs=(select id_produs from produse_v where denumire_produs='Tort cu
fructe');

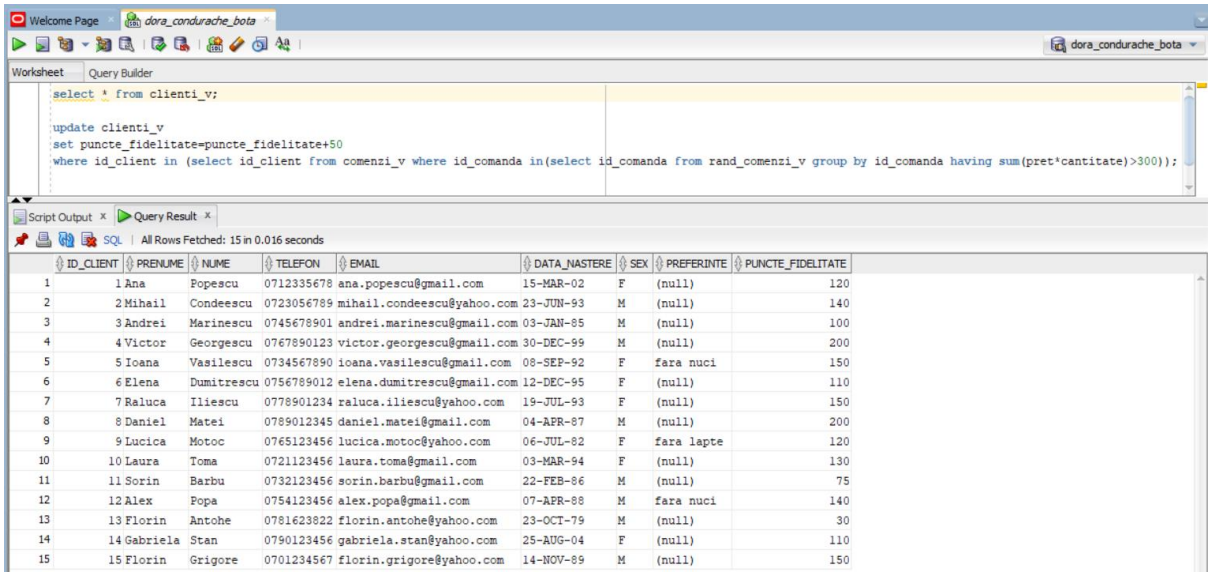
commit;
```



	ID_COMANDA	ID_PRODUS	PRET	CANTITATE
1	1	7	100	3
2	2	12	80	1.2
3	2	3	120	2
4	3	4	90	1.5
5	4	6	90	0.4
6	5	14	85	1
7	6	2	120	2.5
8	6	12	80	1
9	7	8	100	2.5
10	8	4	90	2
11	8	9	120	2
12	9	10	85	0.8
13	9	15	50	0.3
14	10	13	150	0.8
15	11	11	90	2
16	12	9	120	3
17	13	1	100	2.5
18	13	15	50	0.2
19	14	3	120	3
20	15	5	90	2

3) Dăm un bonus de puncte de fidelitate pentru clienții care au plasat comenzi de peste 300 lei (4 rânduri se modifică).

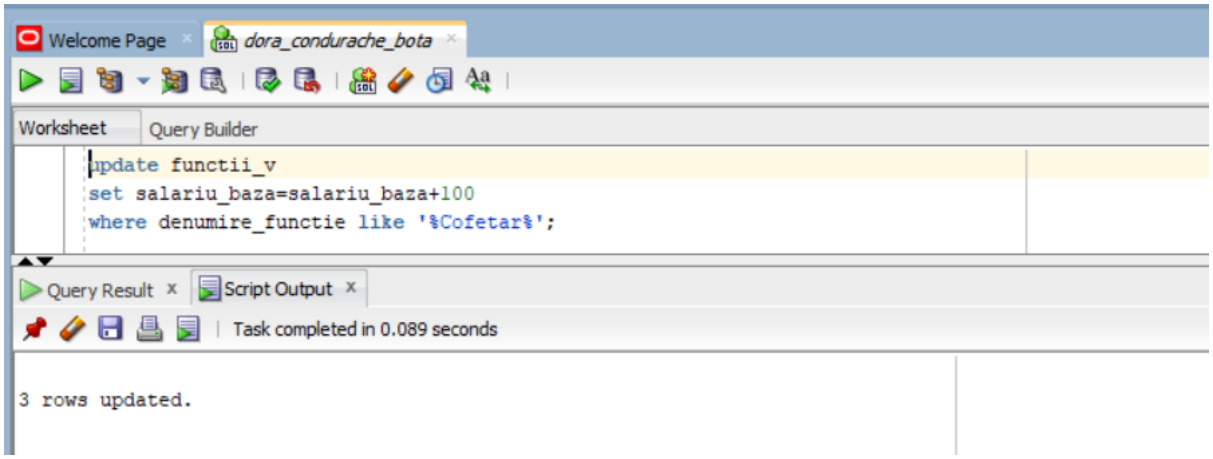
```
update clienti_v
set puncte_fidelitate=puncte_fidelitate+50
where id_client in (select id_client from comenzi_v where id_comanda in(select id_comanda
from rand_comenzi_v group by id_comanda having sum(pret*cantitate)>300));
```

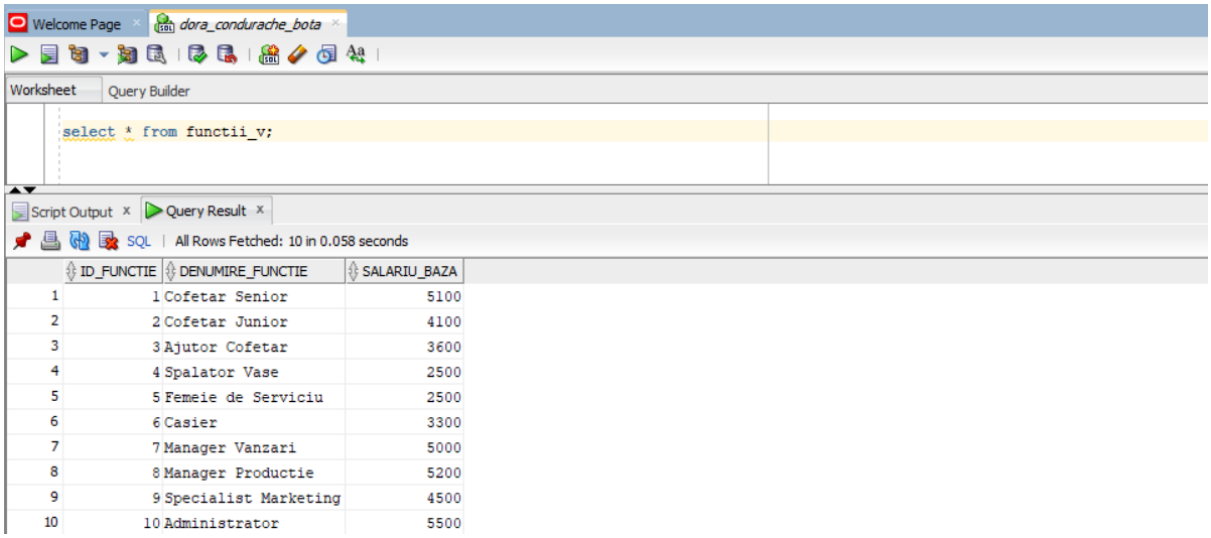


	ID_CLIENT	PRENUME	NUME	TELEFON	EMAIL	DATA_NASTERE	SEX	PREFERINTE	PUNCTE_FIDELITATE
1	1	Ana	Popescu	0712335678	ana.popescu@gmail.com	15-MAR-02	F	(null)	120
2	2	Mihail	Condeescu	0723056789	mihail.condeescu@yahoo.com	23-JUN-93	M	(null)	140
3	3	Andrei	Marinescu	0745678901	andrei.marinescu@gmail.com	03-JAN-85	M	(null)	100
4	4	Victor	Georgescu	0767890123	victor.georgescu@gmail.com	30-DEC-99	M	(null)	200
5	5	Ioana	Vasilescu	0734567890	ioana.vasilescu@gmail.com	08-SEP-92	F	fara nuci	150
6	6	Elena	Dumitrescu	0756789012	elena.dumitrescu@gmail.com	12-DEC-95	F	(null)	110
7	7	Raluca	Iliescu	0778901234	raluca.iliescu@yahoo.com	19-JUL-93	F	(null)	150
8	8	Daniel	Matei	0789012345	daniel.matei@gmail.com	04-APR-87	M	(null)	200
9	9	Lucica	Motoc	0765123456	lucica.motoc@yahoo.com	06-JUL-82	F	fara lapte	120
10	10	Laura	Toma	0721123456	laura.toma@gmail.com	03-MAR-94	F	(null)	130
11	11	Sorin	Barbu	0732123456	sorin.barbu@gmail.com	22-FEB-86	M	(null)	75
12	12	Alex	Popa	0754123456	alex.popa@gmail.com	07-APR-88	M	fara nuci	140
13	13	Florin	Antohe	0781623822	florin.antohe@yahoo.com	23-OCT-79	M	(null)	30
14	14	Gabriela	Stan	0790123456	gabriela.stan@yahoo.com	25-AUG-04	F	(null)	110
15	15	Florin	Grigore	0701234567	florin.grigore@yahoo.com	14-NOV-89	M	(null)	150

4) Creștem salariul de bază al cofetarilor și ajutorilor de cofetari cu 100 lei (se modifică 3 rânduri).

```
update functii_v
set salariu_baza=salariu_baza+100
where denumire_functie like '%Cofetar%';
```





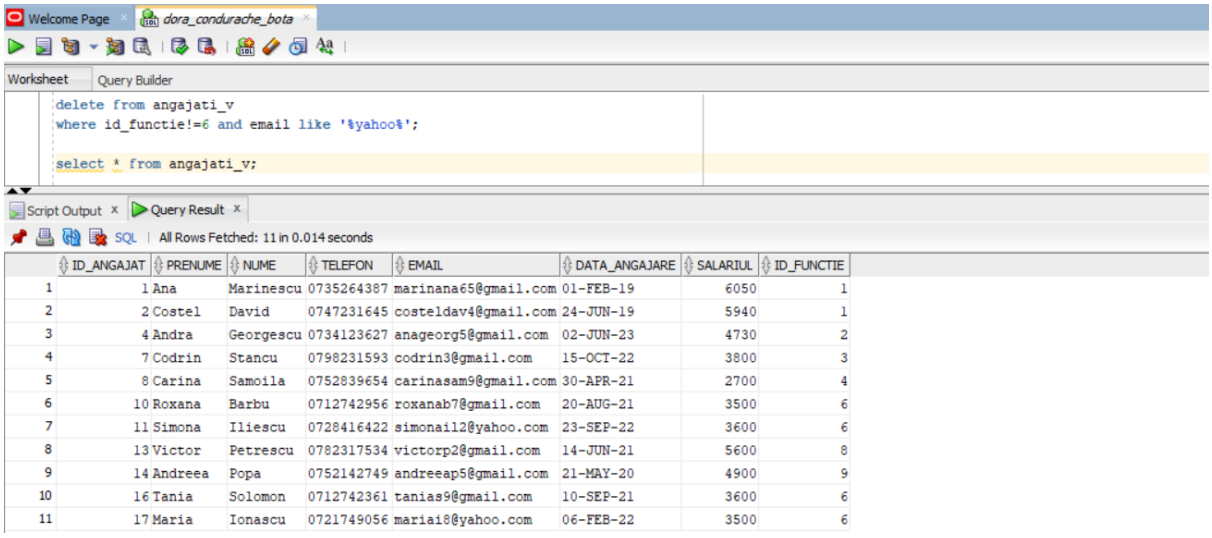
	ID_FUNCTIE	DENUMIRE_FUNCTIE	SALARIU_BAZA
1	1	Cofetar Senior	5100
2	2	Cofetar Junior	4100
3	3	Ajutor Cofetar	3600
4	4	Spalator Vase	2500
5	5	Femeie de Serviciu	2500
6	6	Casier	3300
7	7	Manager Vanzari	5000
8	8	Manager Productie	5200
9	9	Specialist Marketing	4500
10	10	Administrator	5500

5) Ștergem din baza de date angajații care au Yahoo Mail, apoi anulăm tranzacția.

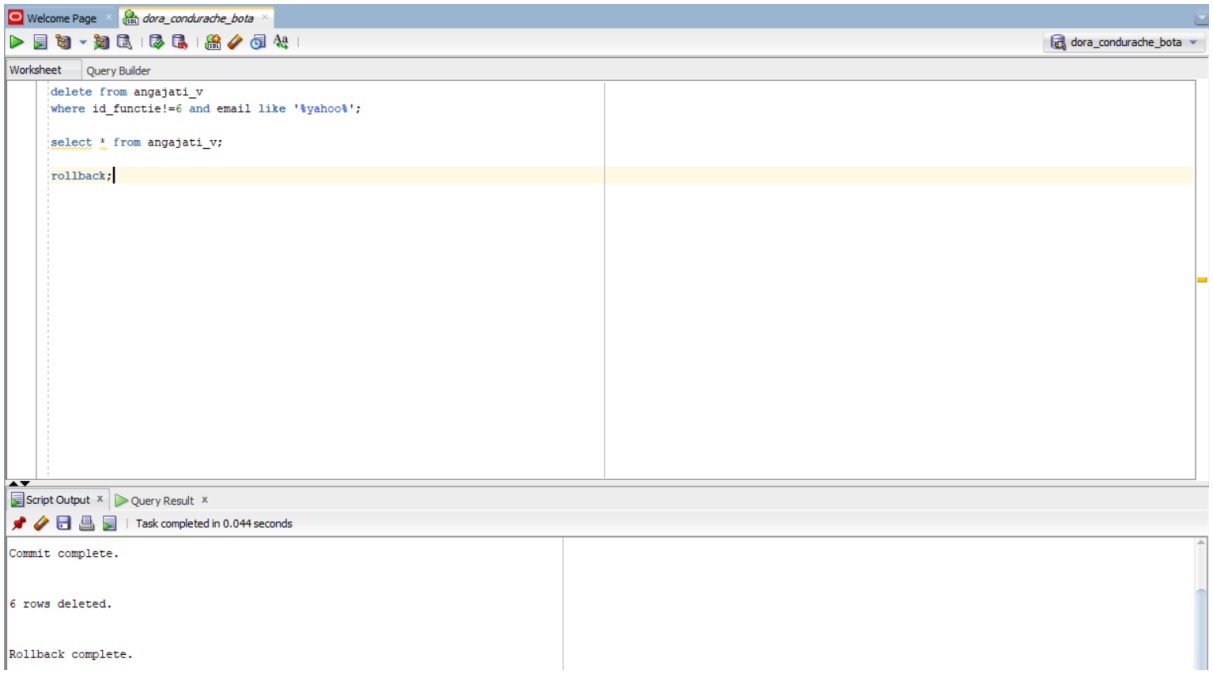
```
delete from angajati_v
where id_functie!=6 and email like '%yahoo%';

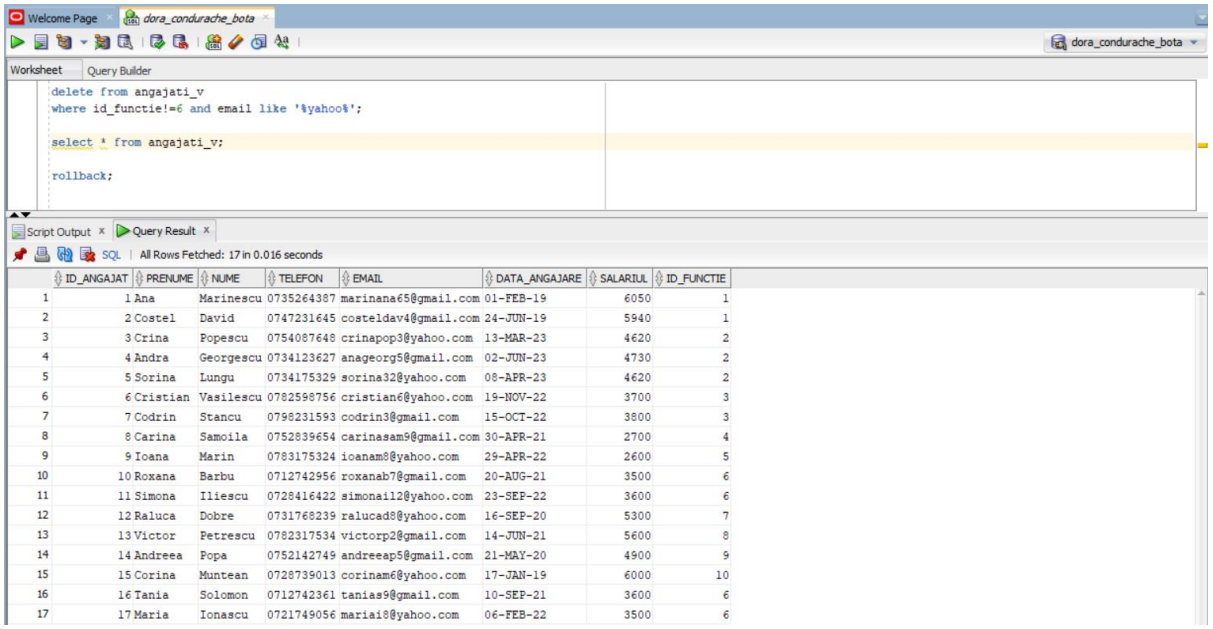
select * from angajati_v;

rollback;
```



	ID_ANGAJAT	PRENUME	NUME	TELEFON	EMAIL	DATA_ANGAJARE	SALARIUL	ID_FUNCTIE
1	1	Ana	Marinescu	0735264387	marinana65@gmail.com	01-FEB-19	6050	1
2	2	Costel	David	0747231645	costeldav4@gmail.com	24-JUN-19	5940	1
3	4	Andra	Georgescu	0734123627	anageorg5@gmail.com	02-JUN-23	4730	2
4	7	Codrin	Stancu	0798231593	codrin3@gmail.com	15-OCT-22	3800	3
5	8	Carina	Samoila	0752839654	carinasam9@gmail.com	30-APR-21	2700	4
6	10	Roxana	Barbu	0712742956	roxanab7@gmail.com	20-AUG-21	3500	6
7	11	Simona	Iliescu	0728416422	simonail2@yahoo.com	23-SEP-22	3600	6
8	13	Victor	Petrescu	0782317534	victorp2@gmail.com	14-JUN-21	5600	8
9	14	Andreea	Popa	0752142749	andreeap5@gmail.com	21-MAY-20	4900	9
10	16	Tania	Solomon	0712742361	tanias9@gmail.com	10-SEP-21	3600	6
11	17	Maria	Ionascu	0721749056	mariai8@yahoo.com	06-FEB-22	3500	6





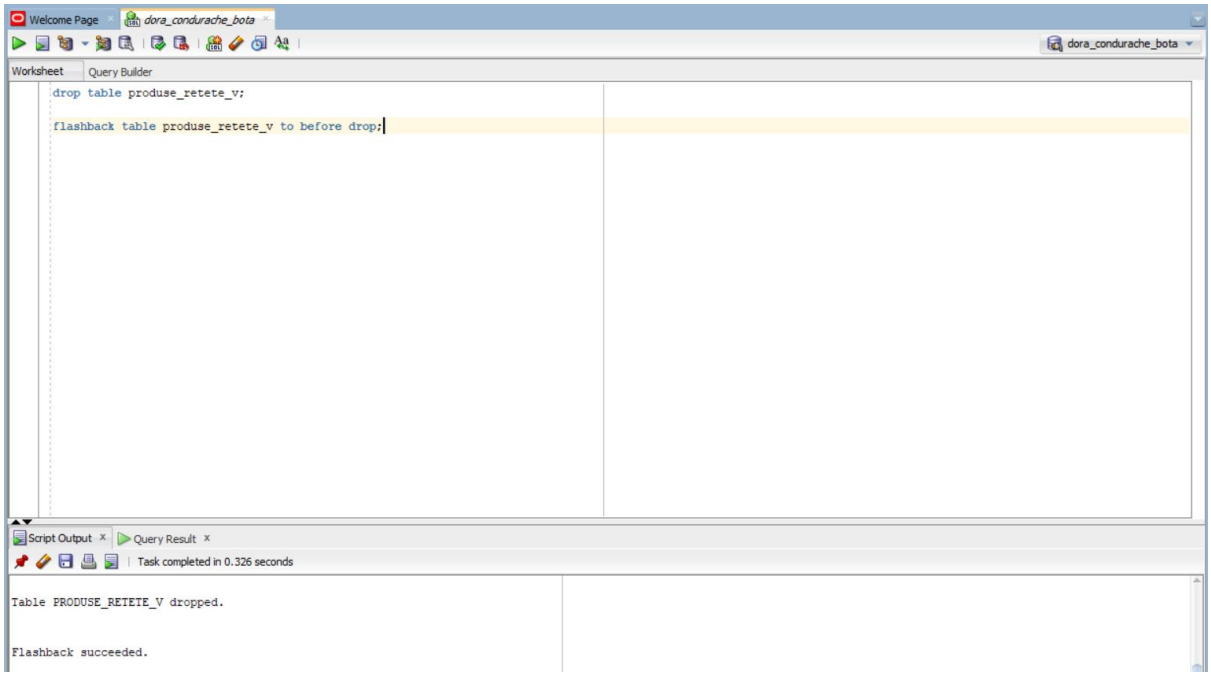
	ID_ANGAJAT	PRENUME	NUME	TELEFON	EMAIL	DATA_ANGAJARE	SALARIUL	ID_FUNCTIE
1	1	Ana	Marinescu	0735264387	marinana65@gmail.com	01-FEB-19	6050	1
2	2	Costel	David	0747231645	costeldav4@gmail.com	24-JUN-19	5940	1
3	3	Crina	Popescu	0754087648	crinapop3@yahoo.com	13-MAR-23	4620	2
4	4	Andra	Georgescu	0734123627	anageorg5@gmail.com	02-JUN-23	4730	2
5	5	Sorina	Lungu	0734175329	sorina32@yahoo.com	08-APR-23	4620	2
6	6	Cristian	Vasilescu	0782598756	cristian6@yahoo.com	19-NOV-22	3700	3
7	7	Codrin	Stancu	0798231593	codrin3@gmail.com	15-OCT-22	3800	3
8	8	Carina	Samoila	0752839654	carinasam9@gmail.com	30-APR-21	2700	4
9	9	Ioana	Marin	0783175324	ioanam8@yahoo.com	29-APR-22	2600	5
10	10	Roxana	Barbu	0712742956	roxanab7@gmail.com	20-AUG-21	3500	6
11	11	Simona	Iliescu	0728416422	simonail2@yahoo.com	23-SEP-22	3600	6
12	12	Raluca	Dobre	0731768239	ralucad8@yahoo.com	16-SEP-20	5300	7
13	13	Victor	Petrescu	0782317534	victorp2@gmail.com	14-JUN-21	5600	8
14	14	Andreea	Popa	0752142749	andreeap5@gmail.com	21-MAY-20	4900	9
15	15	Corina	Muntean	0728739013	corinam6@yahoo.com	17-JAN-19	6000	10
16	16	Tania	Solomon	0712742361	tanias9@gmail.com	10-SEP-21	3600	6
17	17	Maria	Ionascu	0721749056	mariai8@yahoo.com	06-FEB-22	3500	6

7. Ștergerea și recuperarea unei tabele

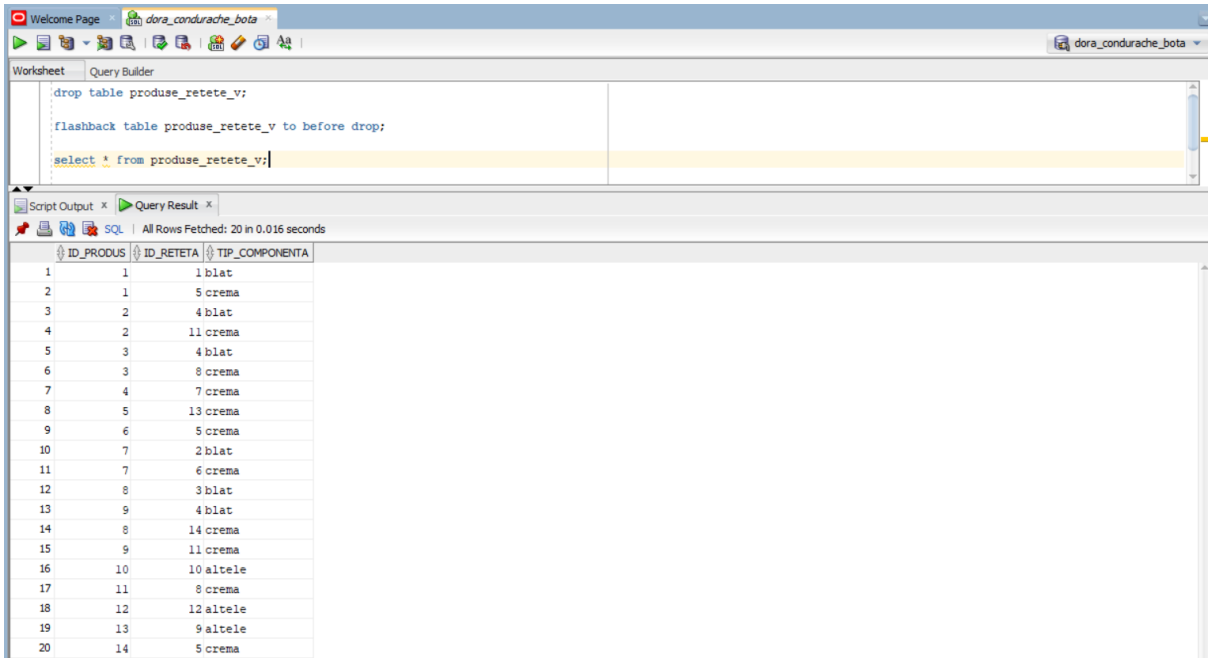
Ștergem tabela Produse_retete_v, apoi o recuperăm.

drop table produse_retete_v;

flashback table produse_retete_v to before drop;



După recuperare, verificăm că tabela există și are toate înregistrările.

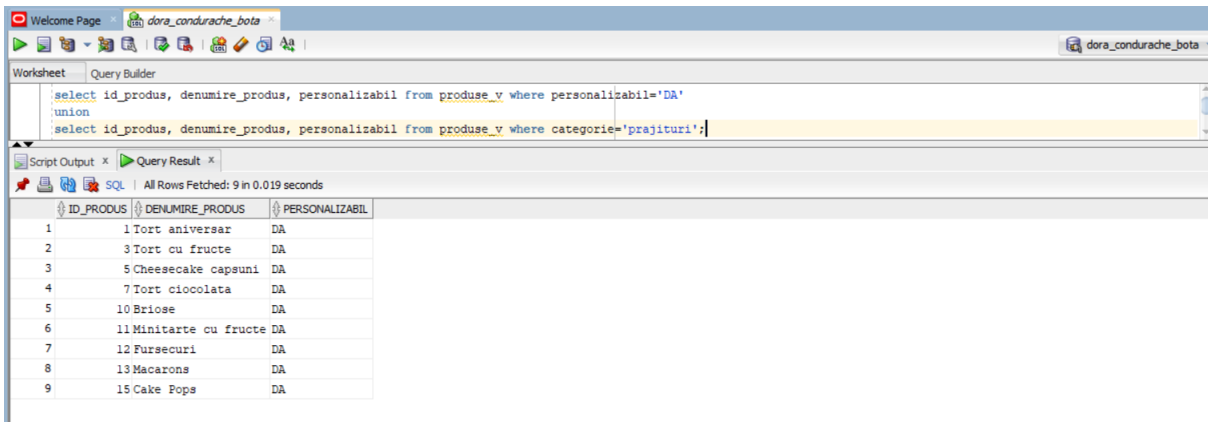


8. Exemple de interogări variate (min 20) – inclunzând şi operatorii UNION, INTERSECT, MINUS, expresiile DECODE şi CASE, cereri imbricate, diverse funcţii single-row, functii de grup, structuri ierarhice, jonctiuni.

1) Să se selecteze produsele care sunt personalizabile sau se găsesc în categoria prăjituri.

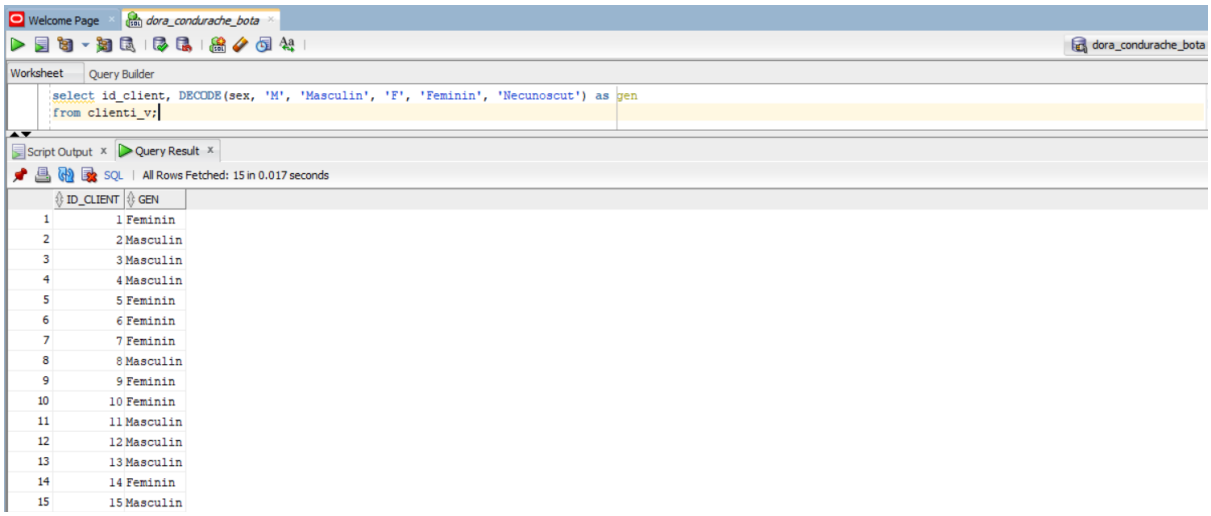
select id_produs, denumire_produs, personalizabil from produse_v where personalizabil='DA'

union
select id_produs, denumire_produs, personalizabil from produse_v where categorie='prajituri';



2) Să se selecteze clienții, scriind genul desfășurat.

select id_client, DECODE(sex, 'M', 'Masculin', 'F', 'Feminin', 'Necunoscut') as gen
from clienti_v;

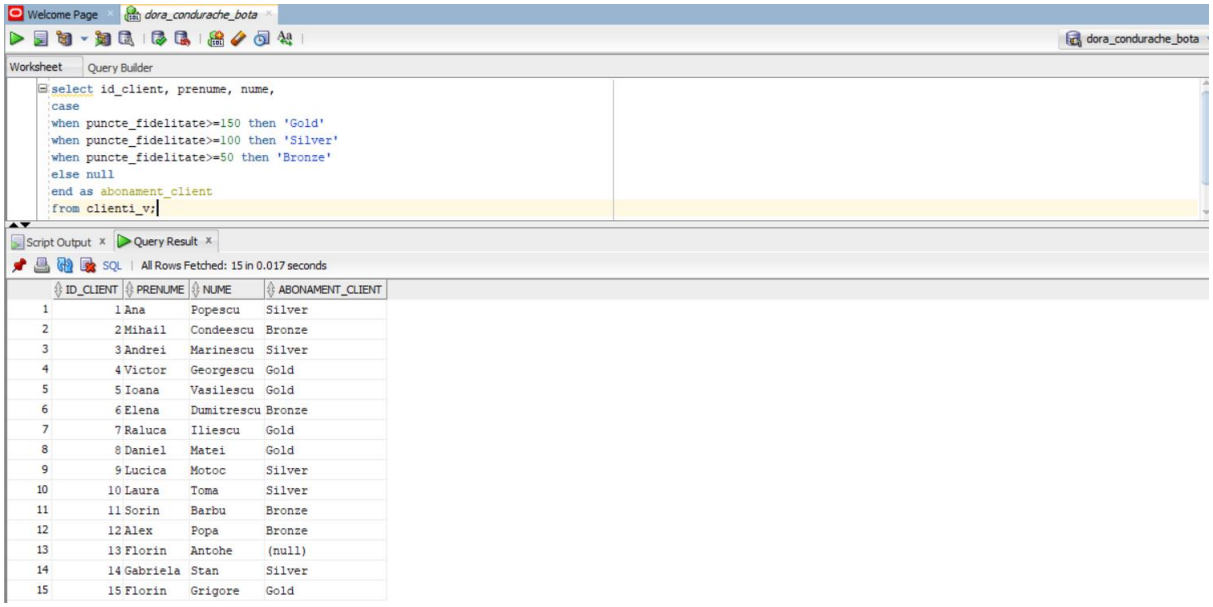


3) Să se ofere abonamente clienților în funcție de punctele de fidelitate:

- Gold dacă punctele sunt mai mari de 150
- Silver dacă punctele sunt mai mari de 100
- Bronze dacă punctele sunt mai mari de 50

select id_client, prenume, nume,
case
when puncte_fidelitate>=150 then 'Gold'
when puncte_fidelitate>=100 then 'Silver'
when puncte_fidelitate>=50 then 'Bronze'
else null

end as abonament_client
from clienti_v;

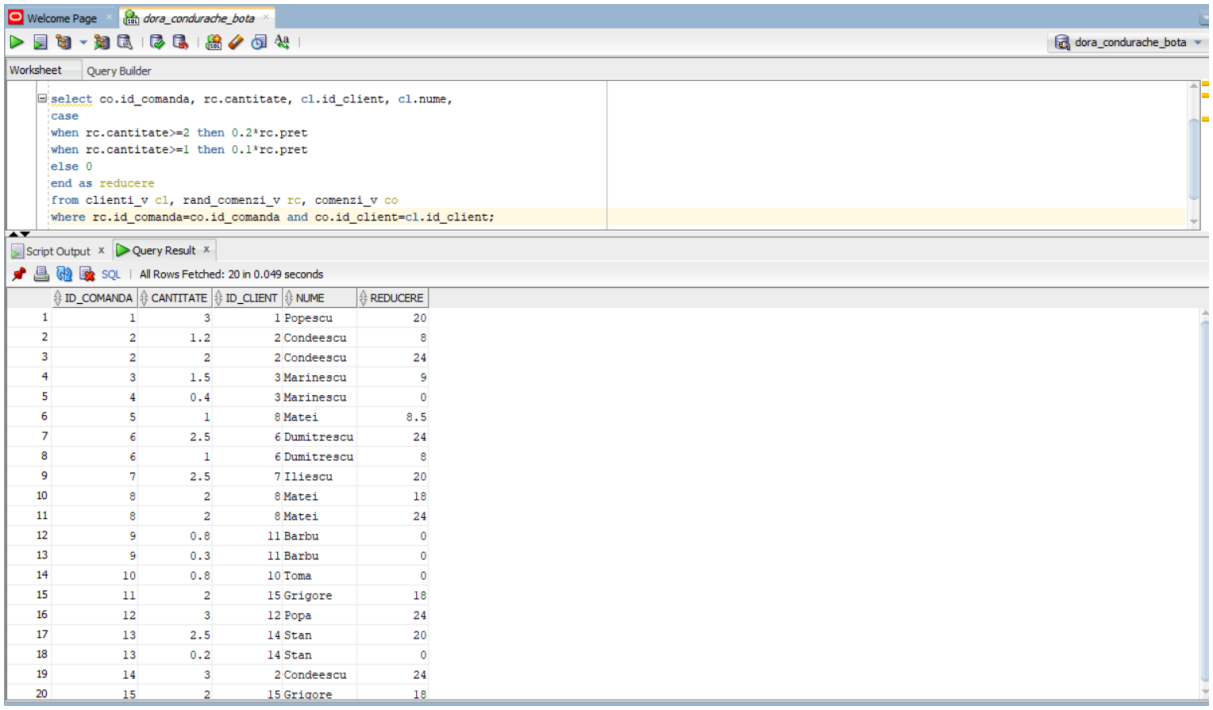


	ID_CLIENT	PRENUME	NUME	ABONAMENT_CLIENT
1	1	Ana	Popescu	Silver
2	2	Mihail	Condeescu	Bronze
3	3	Andrei	Marinescu	Silver
4	4	Victor	Georgescu	Gold
5	5	Ioana	Vasilescu	Gold
6	6	Elena	Dumitrescu	Bronze
7	7	Raluca	Iliescu	Gold
8	8	Daniel	Matei	Gold
9	9	Lucica	Motoc	Silver
10	10	Laura	Toma	Silver
11	11	Sorin	Barbu	Bronze
12	12	Alex	Popa	Bronze
13	13	Florin	Antohe	(null)
14	14	Gabriela	Stan	Silver
15	15	Florin	Grigore	Gold

4) Să se calculeze reducerea pentru produse în functie de cantitatea cumpărată.

- 20% dacă s-a cumpărat mai mult de 2 kg
- 10% dacă s-a cumpărat mai mult de 1 kg
- Nimic altfel.

select co.id_comanda, rc.cantitate, cl.id_client, cl.nume,
case
when rc.cantitate>=2 then 0.2*rc.pret
when rc.cantitate>=1 then 0.1*rc.pret
else 0
end as reducere
from clienti_v cl, rand_comenzi_v rc, comenzi_v co
where rc.id_comanda=co.id_comanda and co.id_client=cl.id_client;



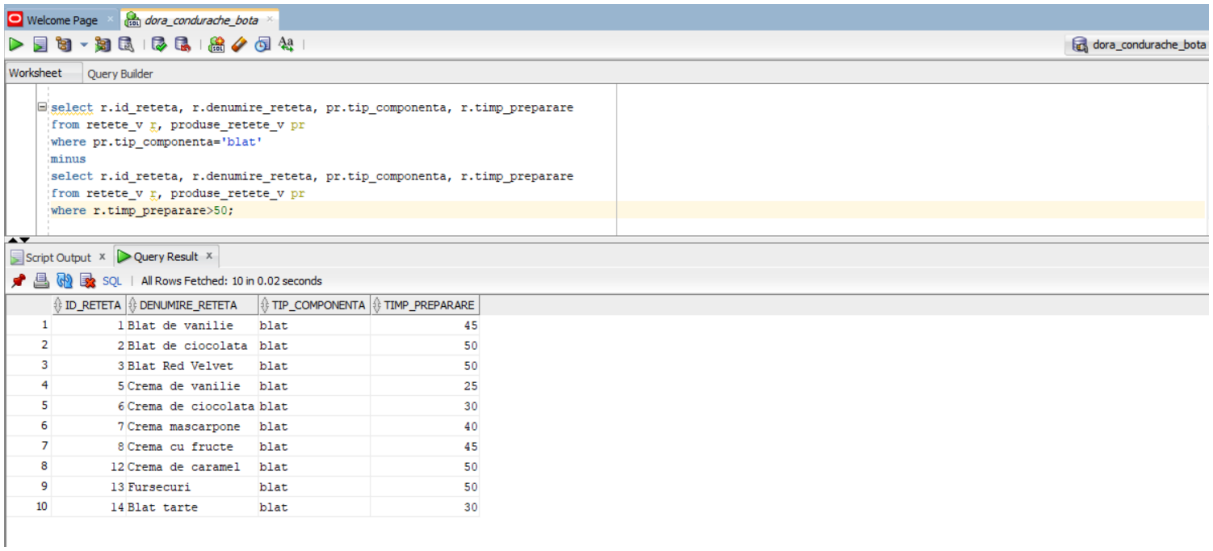
```
select co.id_comanda, rc.cantitate, cl.id_client, cl.nume,
case
when rc.cantitate>=2 then 0.2*rc.pret
when rc.cantitate>=1 then 0.1*rc.pret
else 0
end as reducere
from clienti_v cl, rand_comenzi_v rc, comenzi_v co
where rc.id_comanda=co.id_comanda and co.id_client=cl.id_client;
```

All Rows Fetched: 20 in 0.049 seconds

	ID_COMANDA	CANTITATE	ID_CLIENT	NUME	REDUCERE
1	1	3	1	Popescu	20
2	2	1.2	2	Condeescu	8
3	2	2	2	Condeescu	24
4	3	1.5	3	Marinescu	9
5	4	0.4	3	Marinescu	0
6	5	1	8	Matei	8.5
7	6	2.5	6	Dumitrescu	24
8	6	1	6	Dumitrescu	8
9	7	2.5	7	Iliescu	20
10	8	2	8	Matei	18
11	8	2	8	Matei	24
12	9	0.8	11	Barbu	0
13	9	0.3	11	Barbu	0
14	10	0.8	10	Toma	0
15	11	2	15	Grigore	18
16	12	3	12	Popa	24
17	13	2.5	14	Stan	20
18	13	0.2	14	Stan	0
19	14	3	2	Condeescu	24
20	15	2	15	Grigore	18

5) Să se selecteze rețetele care sunt blaturi și care nu se fac in mai mult de 50 de minute.

```
select r.id_reteta, r.denumire_reteta, pr.tip_componenta, r.timp_preparare
from retete_v r, produse_retete_v pr
where pr.tip_componenta='blat'
minus
select r.id_reteta, r.denumire_reteta, pr.tip_componenta, r.timp_preparare
from retete_v r, produse_retete_v pr
where r.timp_preparare>50;
```



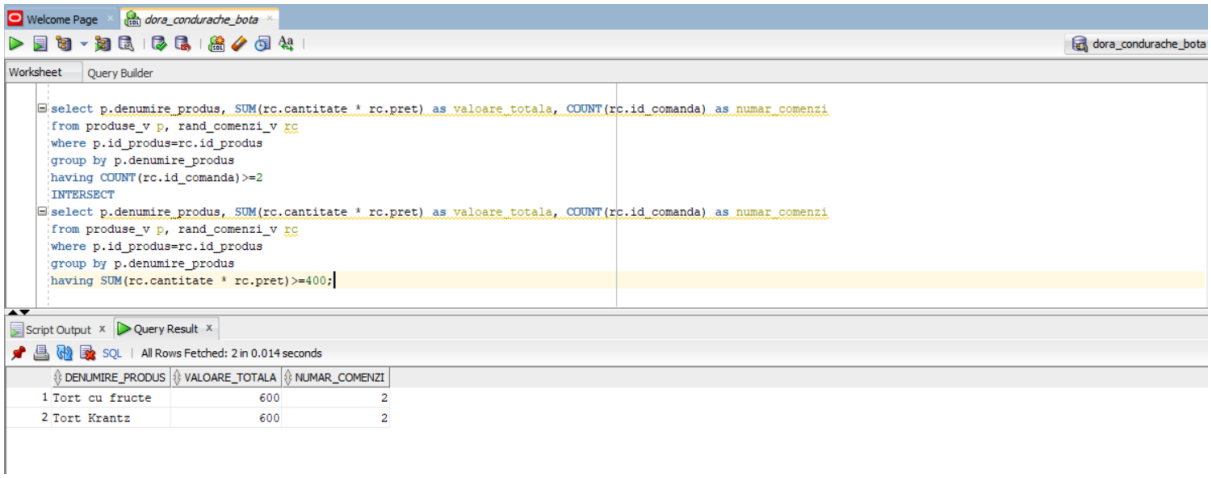
```
select r.id_reteta, r.denumire_reteta, pr.tip_componenta, r.timp_preparare
from retete_v r, produse_retete_v pr
where pr.tip_componenta='blat'
minus
select r.id_reteta, r.denumire_reteta, pr.tip_componenta, r.timp_preparare
from retete_v r, produse_retete_v pr
where r.timp_preparare>50;
```

All Rows Fetched: 10 in 0.02 seconds

	ID_RETETA	DENUMIRE_RETETA	TIP_COMPONENTA	TIMP_PREPARARE
1	1	Blat de vanilie	blat	45
2	2	Blat de ciocolata	blat	50
3	3	Blat Red Velvet	blat	50
4	5	Crema de vanilie	blat	25
5	6	Crema de ciocolata	blat	30
6	7	Crema mascarpone	blat	40
7	8	Crema cu fructe	blat	45
8	12	Crema de caramel	blat	50
9	13	Fursecuri	blat	50
10	14	Blat tarte	blat	30

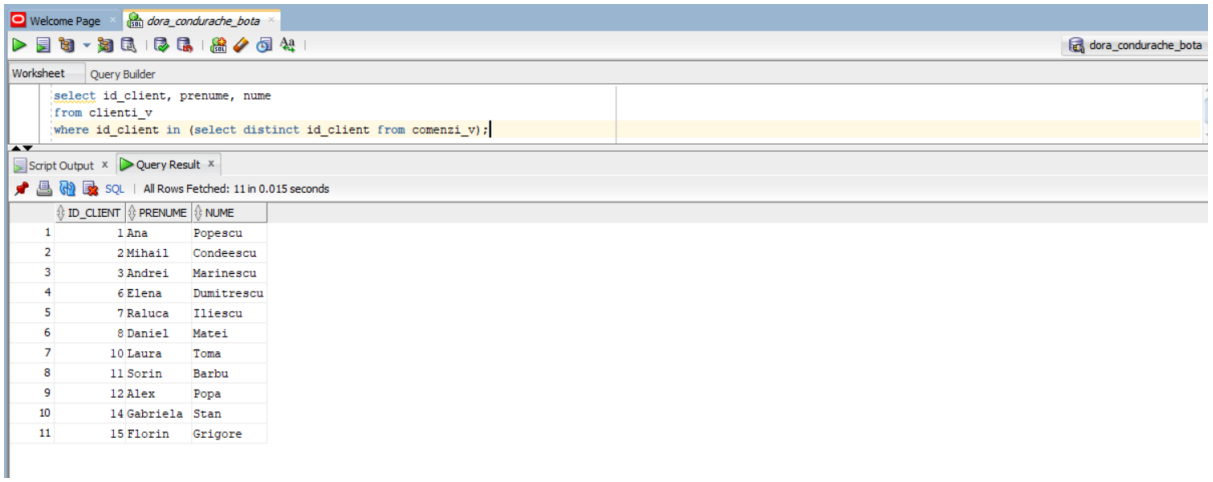
6) Să se selecteze denumirea produselor, valoarea totală a comenzii (SUM(cantitate*pret)) și numărul de comenzi (count(id_comanda)) pentru produsele care au fost comandate cel puțin de două ori și care au valoarea totală a comenzii mai mare de 400 lei.

```
select p.denumire_produs, SUM(rc.cantitate * rc.pret) as valoare_totala,
COUNT(rc.id_comanda) as numar_comenzi
from produse_v p, rand_comenzi_v rc
where p.id_produs=rc.id_produs
group by p.denumire_produs
having COUNT(rc.id_comanda)>=2
INTERSECT
select p.denumire_produs, SUM(rc.cantitate * rc.pret) as valoare_totala,
COUNT(rc.id_comanda) as numar_comenzi
from produse_v p, rand_comenzi_v rc
where p.id_produs=rc.id_produs
group by p.denumire_produs
having SUM(rc.cantitate * rc.pret)>=400;
```



7) Să se selecteze clienții care au plasat cel puțin o comandă.

```
select id_client, prenume, nume
from clienti_v
where id_client in (select distinct id_client from comenzi_v);
```



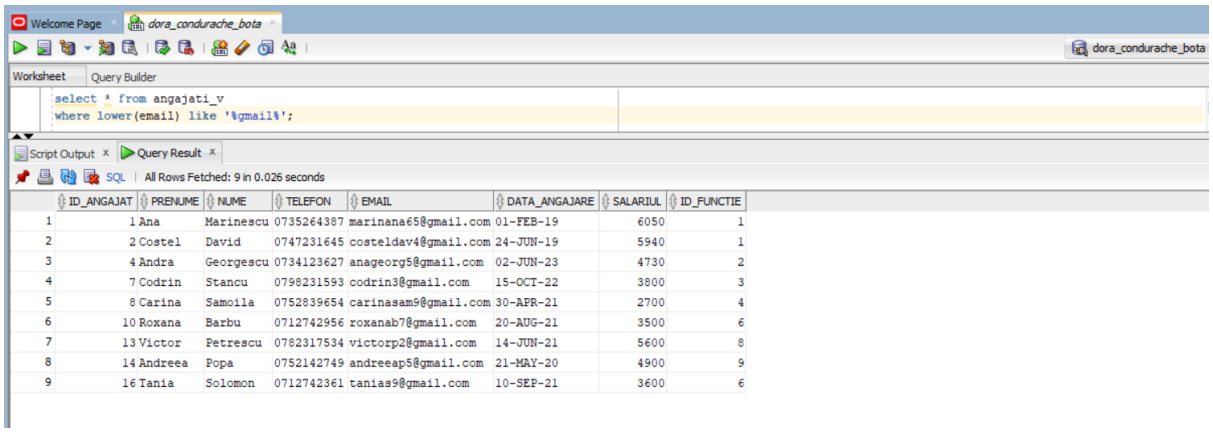
```
select id_client, prenume, nume
from clienti_v
where id_client in (select distinct id_client from comenzi_v);
```

All Rows Fetched: 11 in 0.015 seconds

	ID_CLIENT	PRENUME	NUME
1	1	Ana	Popescu
2	2	Mihail	Condeescu
3	3	Andrei	Marinescu
4	6	Elena	Dumitrescu
5	7	Raluca	Iliescu
6	8	Daniel	Matei
7	10	Laura	Toma
8	11	Sorin	Barbu
9	12	Alex	Popa
10	14	Gabriela	Stan
11	15	Florin	Grigore

8) Să se selecteze angajații care au email la Google.

```
select * from angajati_v
where lower(email) like '%gmail%';
```



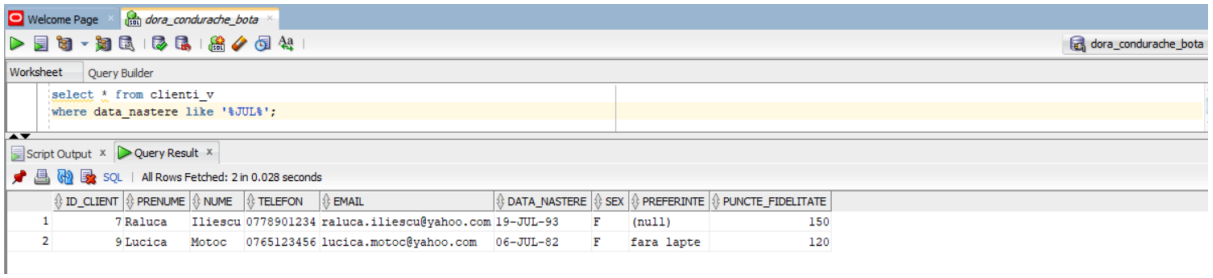
```
select * from angajati_v
where lower(email) like '%gmail%';
```

All Rows Fetched: 9 in 0.026 seconds

	ID_ANGAJAT	PRENUME	NUME	TELEFON	EMAIL	DATA_ANGAJARE	SALARIUL	ID_FUNCTIE
1	1	Ana	Marinescu	0735264387	marinana65@gmail.com	01-FEB-19	6050	1
2	2	Costel	David	0747231645	costeldav4@gmail.com	24-JUN-19	5940	1
3	4	Andra	Georgescu	0734123627	anageorg5@gmail.com	02-JUN-23	4730	2
4	7	Codrin	Stancu	0798231593	codrin3@gmail.com	15-OCT-22	3800	3
5	8	Carina	Samoila	0752839654	carinasam9@gmail.com	30-APR-21	2700	4
6	10	Roxana	Barbu	0712742956	roxanab7@gmail.com	20-AUG-21	3500	6
7	13	Victor	Petrescu	0782317534	victorp2@gmail.com	14-JUN-21	5600	8
8	14	Andreea	Popa	0752142749	andreeap5@gmail.com	21-MAY-20	4900	9
9	16	Tania	Solomon	0712742361	tanias9@gmail.com	10-SEP-21	3600	6

9) Să se selecteze clienții născuți în luna iulie.

```
select * from clienti_v
where data_nastere like '%JUL%';
```



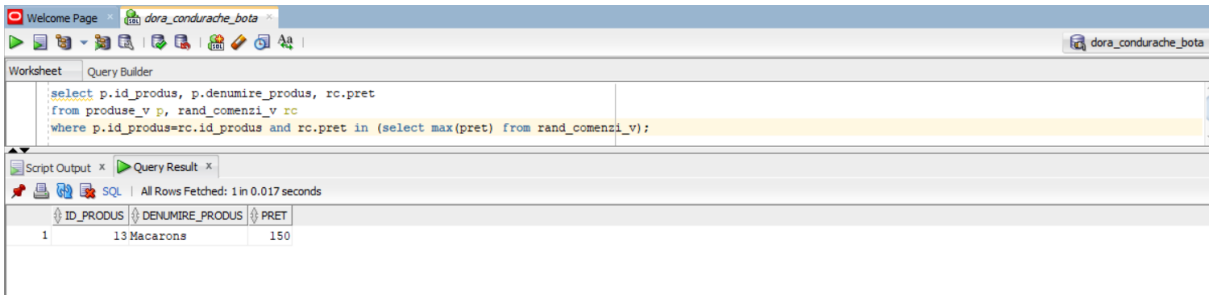
```
select * from clienti_v
where data_nastere like '%JUL%';
```

All Rows Fetched: 2 in 0.028 seconds

	ID_CLIENT	PRENUME	NUME	TELEFON	EMAIL	DATA_NASTERE	SEX	PREFERINTE	PUNCTE_FIDELITATE
1	7	Raluca	Iliescu	0778901234	raluca.iliescu@yahoo.com	19-JUL-93	F	(null)	150
2	9	Lucica	Motoc	0765123456	lucica.motoc@yahoo.com	06-JUL-82	F	fara lapte	120

10) Să se selecteze produsele cu prețul maxim.

```
select p.id_produs, p.denumire_produs, rc.pret
from produse_v p, rand_comenzi_v rc
where p.id_produs=rc.id_produs and rc.pret in (select max(pret) from rand_comenzi_v);
```



	ID_PRODUS	DENUMIRE_PRODUS	PRET
1	13	Macarons	150

11) Să se selecteze toate comenzile care au fost lansate online după 14 noiembrie 2024.

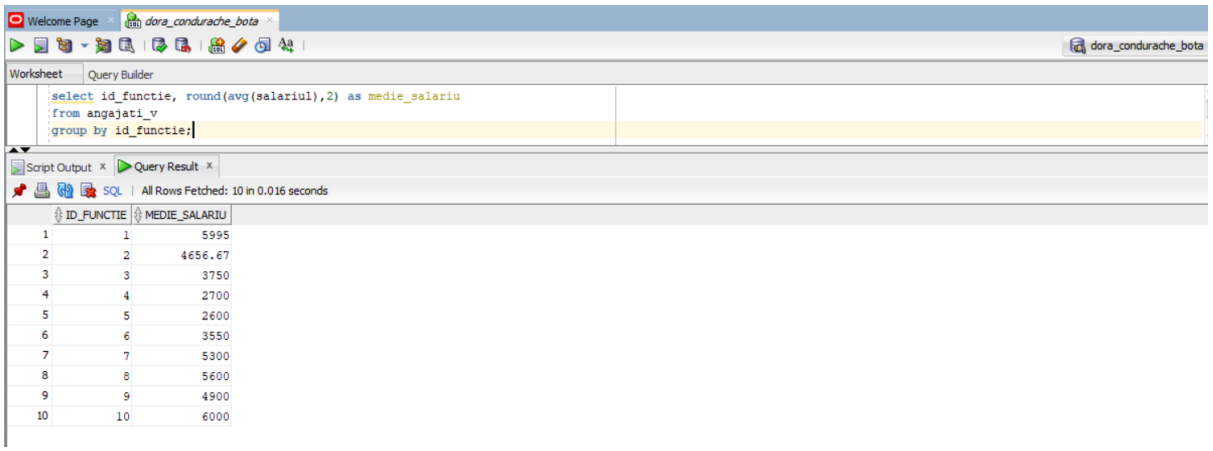
```
select * from comenzi_v
where modalitate='online' and data>to_date('14-NOV-2024', 'DD-MON-YYYY');
```



	ID_COMANDA	DATA	MODALITATE	ID_CLIENT	ID_ANGAJAT
1	8	18-NOV-24	online	8	16
2	12	28-NOV-24	online	12	16
3	14	02-DEC-24	online	2	10

12) Calculați media salariilor pentru fiecare funcție. Folosiți doar 2 zecimale.

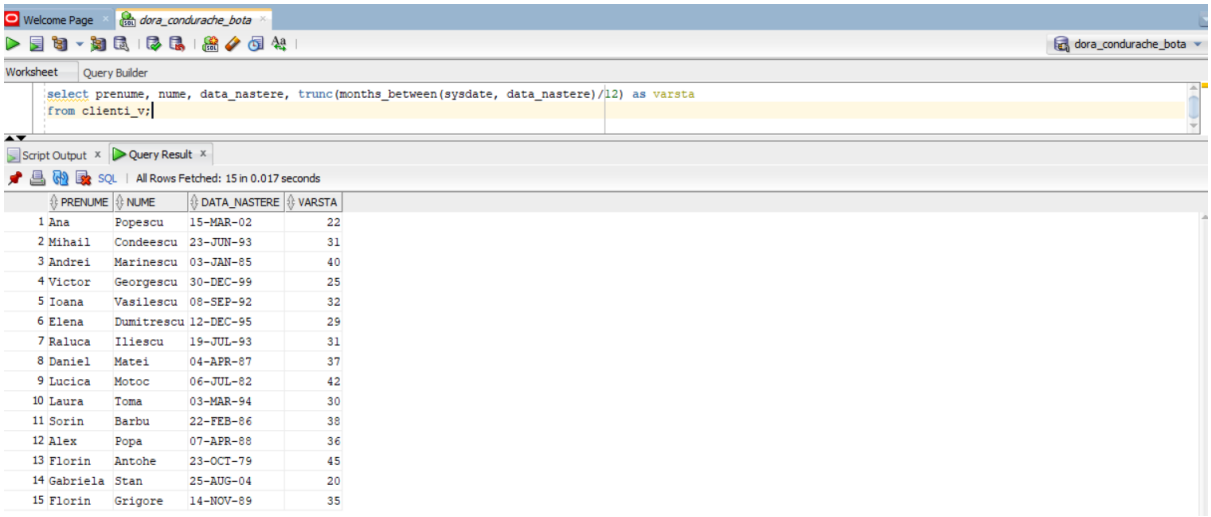
```
select id_functie, round(avg(salariul),2) as medie_salariu
from angajati_v
group by id_functie;
```



	ID_FUNCTIE	MEDIE_SALARIU
1	1	5995
2	2	4656.67
3	3	3750
4	4	2700
5	5	2600
6	6	3550
7	7	5300
8	8	5600
9	9	4900
10	10	6000

13) Calculați vârsta aproximativă în ani a clienților.

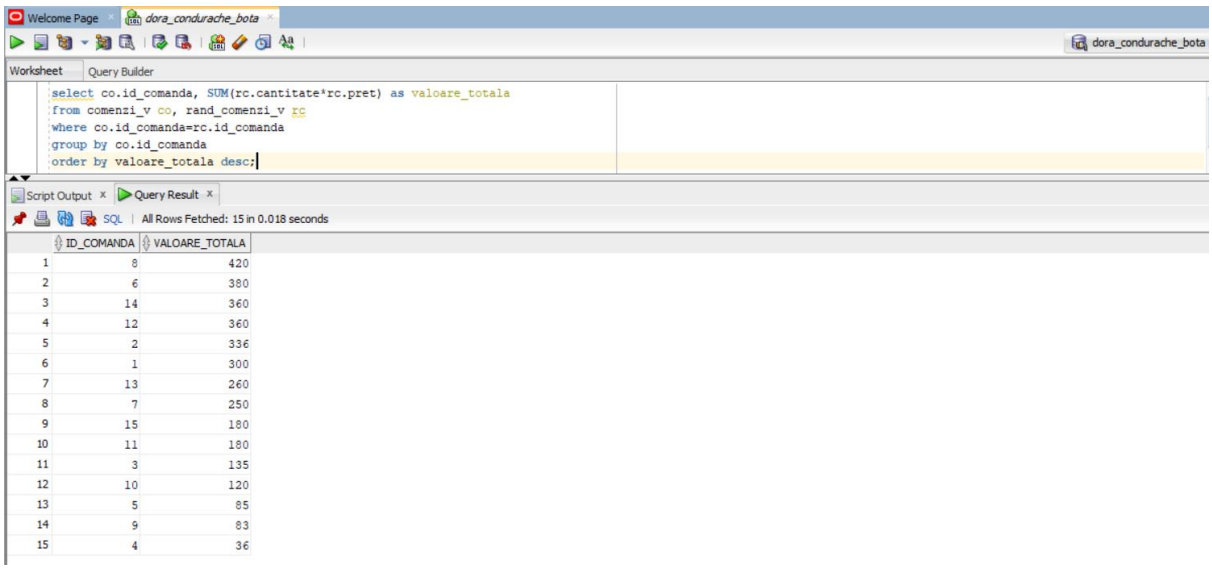
```
select prenume, nume, data_nastere, trunc(months_between(sysdate, data_nastere)/12) as
varsta
from clienti_v;
```



	PRENUME	NUME	DATA_NASTERE	VARSTA
1	Ana	Popescu	15-MAR-02	22
2	Mihail	Condeescu	23-JUN-93	31
3	Andrei	Marinescu	03-JAN-85	40
4	Victor	Georgescu	30-DEC-99	25
5	Ioana	Vasilescu	08-SEP-92	32
6	Elena	Dumitrescu	12-DEC-95	29
7	Raluca	Iliescu	19-JUL-93	31
8	Daniel	Matei	04-APR-87	37
9	Lucica	Motoc	06-JUL-82	42
10	Laura	Toma	03-MAR-94	30
11	Sorin	Barbu	22-FEB-86	38
12	Alex	Popa	07-APR-88	36
13	Florin	Antohe	23-OCT-79	45
14	Gabriela	Stan	25-AUG-04	20
15	Florin	Grigore	14-NOV-89	35

14) Să se calculeze valoarea totală a fiecărei comenzi și să se ordoneze descrescător în funcție de valoarea totală.

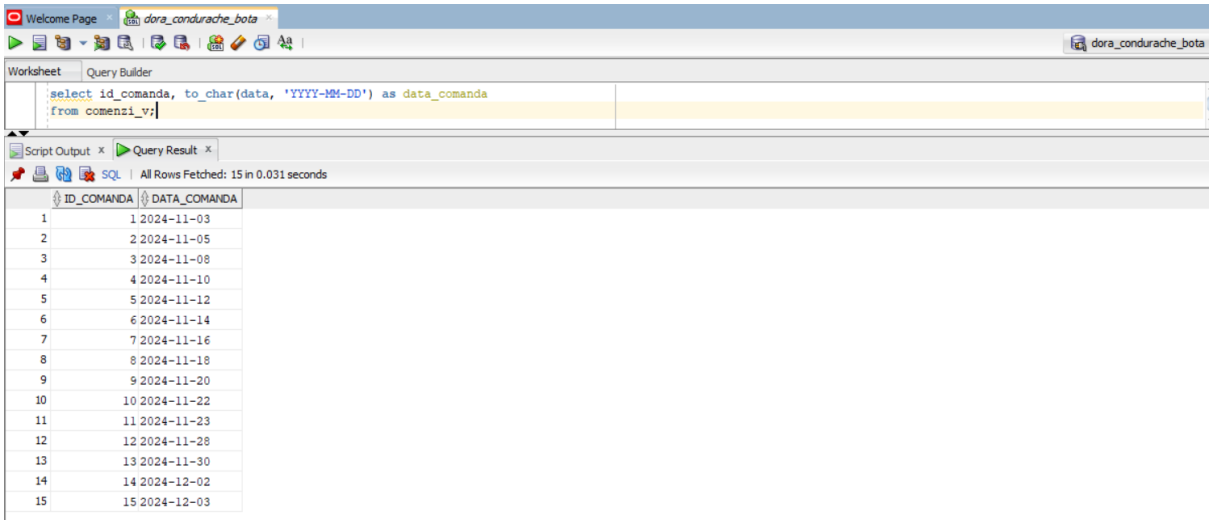
```
select co.id_comanda, SUM(rc.cantitate*rc.pret) as valoare_totala
from comenzi_v co, rand_comenzi_v rc
where co.id_comanda=rc.id_comanda
group by co.id_comanda
order by valoare_totala desc;
```



	ID_COMANDA	VALOARE_TOTALA
1	8	420
2	6	380
3	14	360
4	12	360
5	2	336
6	1	300
7	13	260
8	7	250
9	15	180
10	11	180
11	3	135
12	10	120
13	5	85
14	9	83
15	4	36

15) Afișați data comenzilor sub un alt format.

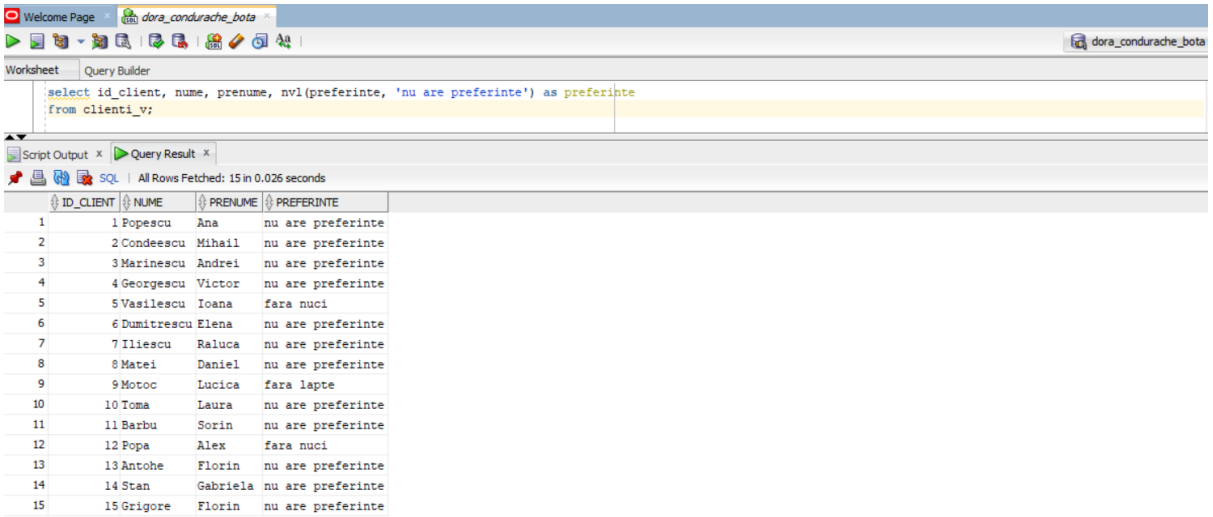
```
select id_comanda, to_char(data, 'YYYY-MM-DD') as data_comanda
from comenzi_v;
```



	ID_COMANDA	DATA_COMANDA
1	1	2024-11-03
2	2	2024-11-05
3	3	2024-11-08
4	4	2024-11-10
5	5	2024-11-12
6	6	2024-11-14
7	7	2024-11-16
8	8	2024-11-18
9	9	2024-11-20
10	10	2024-11-22
11	11	2024-11-23
12	12	2024-11-28
13	13	2024-11-30
14	14	2024-12-02
15	15	2024-12-03

16) Să se înlocuiască valoarea NULL la clienții fără preferințe.

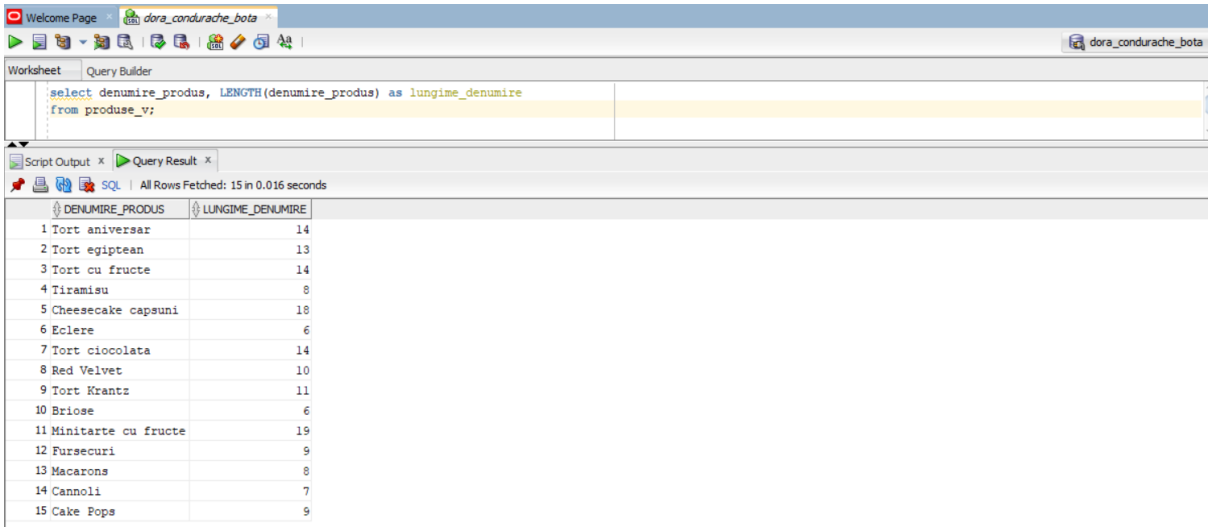
```
select id_client, nume, prenume, nvl(preferinte, 'nu are preferinte') as preferinte
from clienti_v;
```



	ID_CLIENT	NUME	PRENUME	PREFERINTE
1	1	Popescu	Ana	nu are preferinte
2	2	Condeescu	Mihail	nu are preferinte
3	3	Marinescu	Andrei	nu are preferinte
4	4	Georgescu	Victor	nu are preferinte
5	5	Vasilescu	Ioana	fara nuci
6	6	Dumitrescu	Elena	nu are preferinte
7	7	Iliescu	Raluca	nu are preferinte
8	8	Matei	Daniel	nu are preferinte
9	9	Motoc	Lucica	fara lapte
10	10	Toma	Laura	nu are preferinte
11	11	Barbu	Sorin	nu are preferinte
12	12	Popa	Alex	fara nuci
13	13	Antohe	Florin	nu are preferinte
14	14	Stan	Gabriela	nu are preferinte
15	15	Grigore	Florin	nu are preferinte

17) Să se calculeze lungimea denumirii fiecărui produs.

```
select denumire_produs, LENGTH(denumire_produs) as lungime_denumire
from produse_v;
```



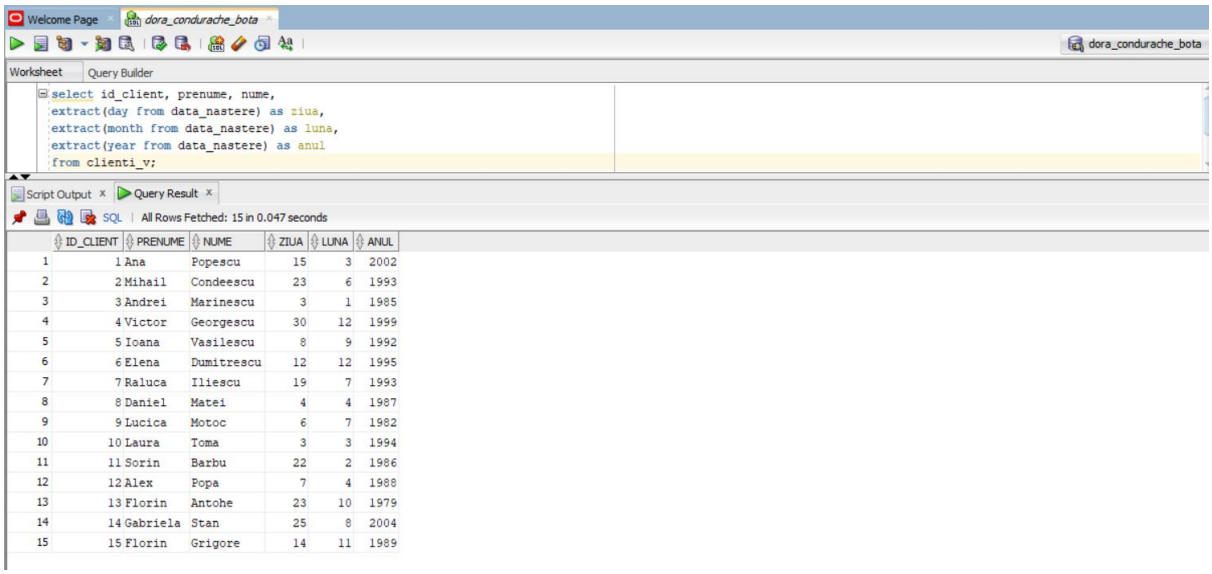
```
select denumire_produs, LENGTH(denumire_produs) as lungime_denumire
from produse_v;
```

All Rows Fetched: 15 in 0.016 seconds

	DENUMIRE_PRODUS	LUNGIME_DENUMIRE
1	Tort aniversar	14
2	Tort egiptean	13
3	Tort cu fructe	14
4	Tiramisu	8
5	Cheesecake capsuni	18
6	Eclere	6
7	Tort ciocolata	14
8	Red Velvet	10
9	Tort Krantz	11
10	Briose	6
11	Minitarte cu fructe	19
12	Fursecuri	9
13	Macarons	8
14	Cannoli	7
15	Cake Pops	9

18) Să se afișeze separat ziua, luna și anul nașterii clienților.

```
select id_client, prenume, nume,
extract(day from data_nastere) as ziua,
extract(month from data_nastere) as luna,
extract(year from data_nastere) as anul
from clienti_v;
```



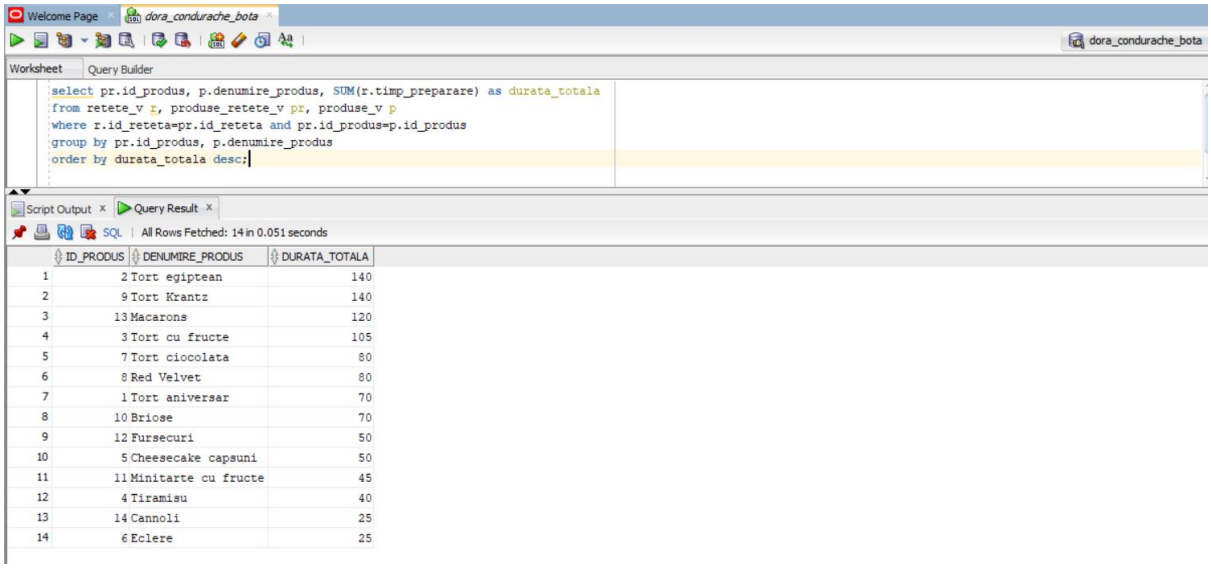
```
select id_client, prenume, nume,
extract(day from data_nastere) as ziua,
extract(month from data_nastere) as luna,
extract(year from data_nastere) as anul
from clienti_v;
```

All Rows Fetched: 15 in 0.047 seconds

	ID_CLIENT	PRENUME	NUME	ZIUA	LUNA	ANUL
1	1	Ana	Popescu	15	3	2002
2	2	Mihail	Condeescu	23	6	1993
3	3	Andrei	Marinescu	3	1	1985
4	4	Victor	Georgescu	30	12	1999
5	5	Ioana	Vasilescu	8	9	1992
6	6	Elena	Dumitrescu	12	12	1995
7	7	Raluca	Iliescu	19	7	1993
8	8	Daniel	Matei	4	4	1987
9	9	Lucica	Motoc	6	7	1982
10	10	Laura	Toma	3	3	1994
11	11	Sorin	Barbu	22	2	1986
12	12	Alex	Popa	7	4	1988
13	13	Florin	Antohe	23	10	1979
14	14	Gabriela	Stan	25	8	2004
15	15	Florin	Grigore	14	11	1989

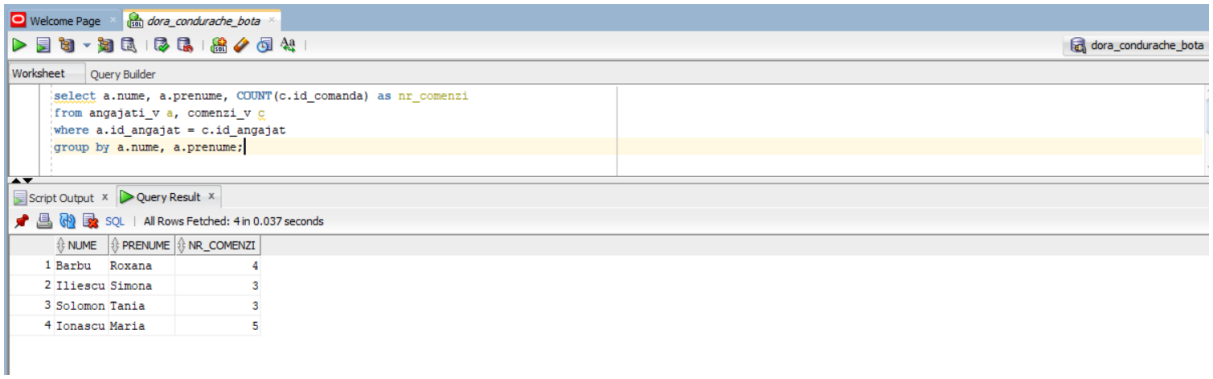
19) Să se calculeze durata totală de preparare a fiecărui produs.

```
select pr.id_produs, p.denumire_produs, SUM(r.timp_preparare) as durata_totala
from retete_v r, produse_retete_v pr, produse_v p
where r.id_reteta=pr.id_reteta and pr.id_produs=p.id_produs
group by pr.id_produs, p.denumire_produs
order by durata_totala desc;
```



20) Selectați angajații și numărul de comenzi gestionate de ei.

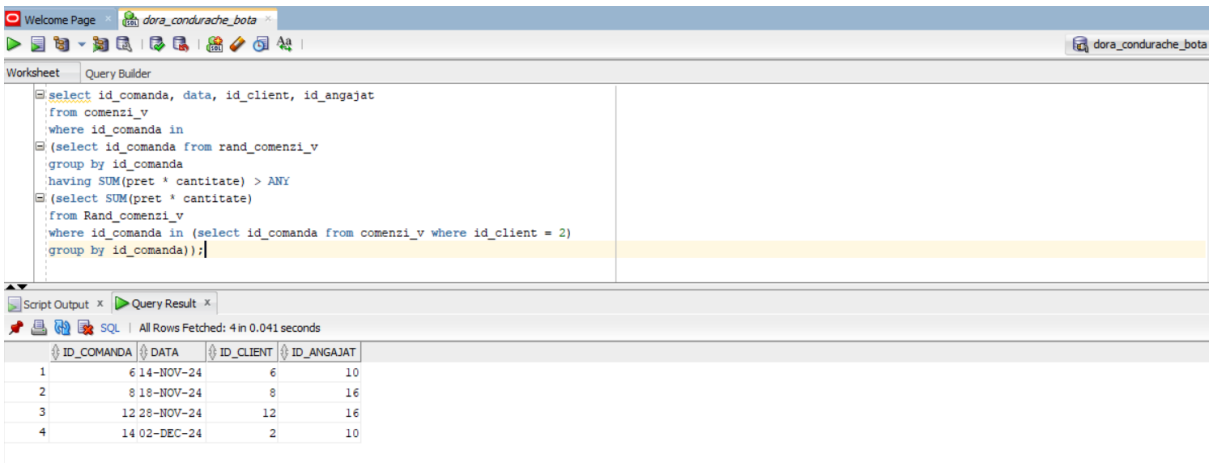
```
select a.nume, a.prenume, COUNT(c.id_comanda) as nr_comenzi
from angajati_v a, comenzi_v c
where a.id_angajat = c.id_angajat
group by a.nume, a.prenume;
```



21) Să se selecteze comenzile care au o valoare mai mare decât oricare dintre comenzile plasate de clientul cu id_client=2.

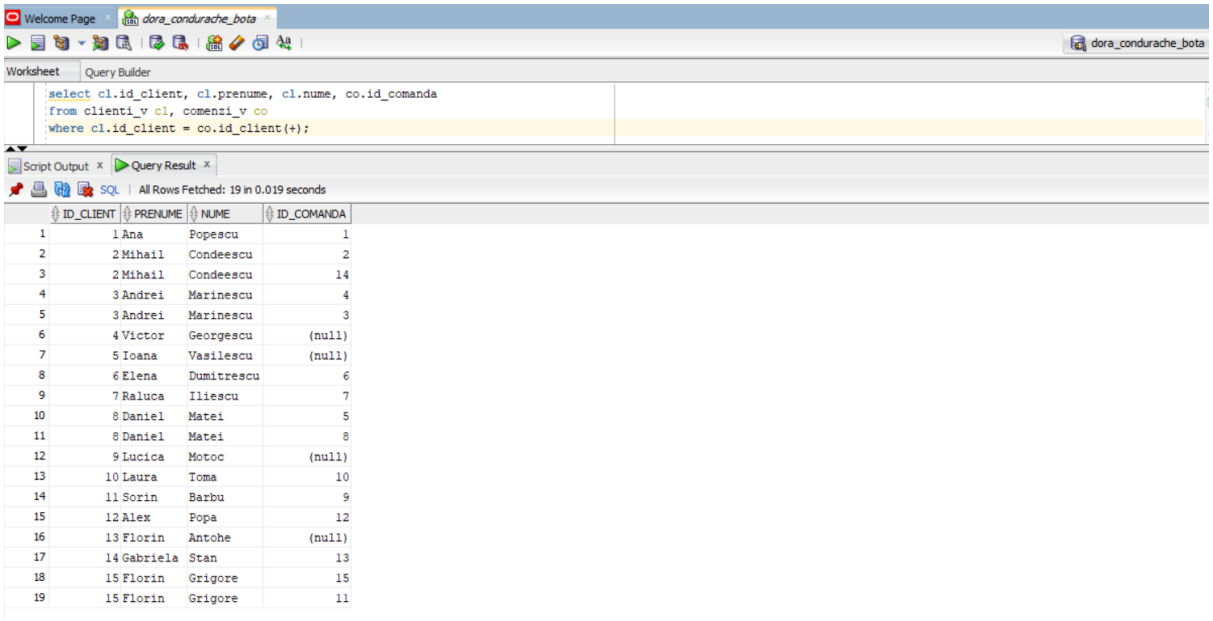
```
select id_comanda, data, id_client, id_angajat
from comenzi_v
where id_comanda in
(select id_comanda from rand_comenzi_v
group by id_comanda
having SUM(pret * cantitate) > ANY
(select SUM(pret * cantitate)
from Rand_comenzi_v
```

where id_comanda in (select id_comanda from comenzi_v where id_client = 2)
group by id_comanda));



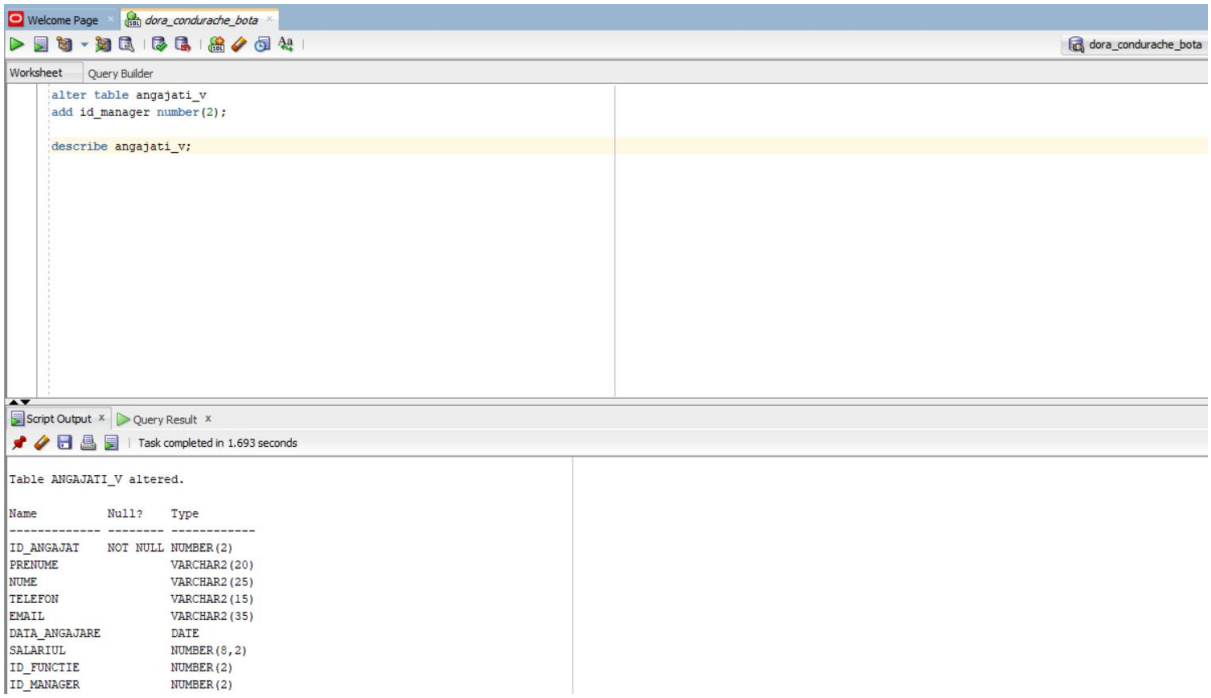
22) Să se afișeze clienții și comenzile lor, indiferent dacă au plasat o comandă sau nu.

select cl.id_client, cl.prenume, cl.nume, co.id_comanda
from clienti_v cl, comenzi_v co
where cl.id_client = co.id_client(+);

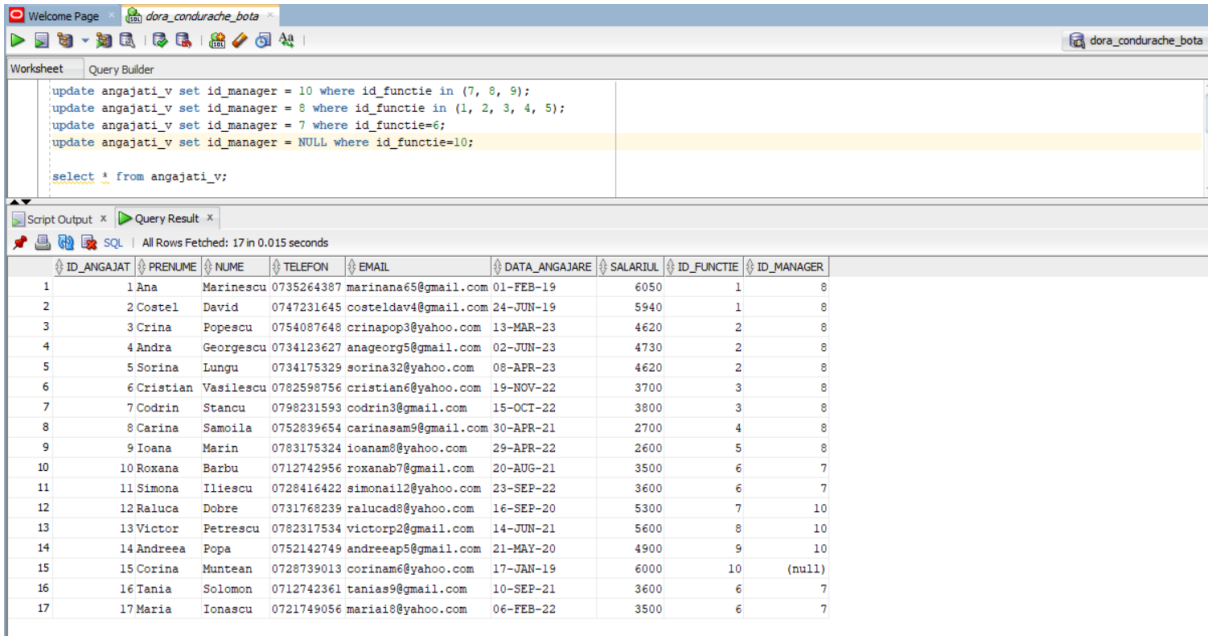


23) Adăugăm coloana id_manager în tabela Angajati_v și înregistrările pentru a putea lucra cu structuri ierarhice.

alter table angajati_v
add id_manager number(2);



update angajati_v set id_manager = 10 where id_functie in (7, 8, 9);
update angajati_v set id_manager = 8 where id_functie in (1, 2, 3, 4, 5);
update angajati_v set id_manager = 7 where id_functie=6;
update angajati_v set id_manager = NULL where id_functie=10;

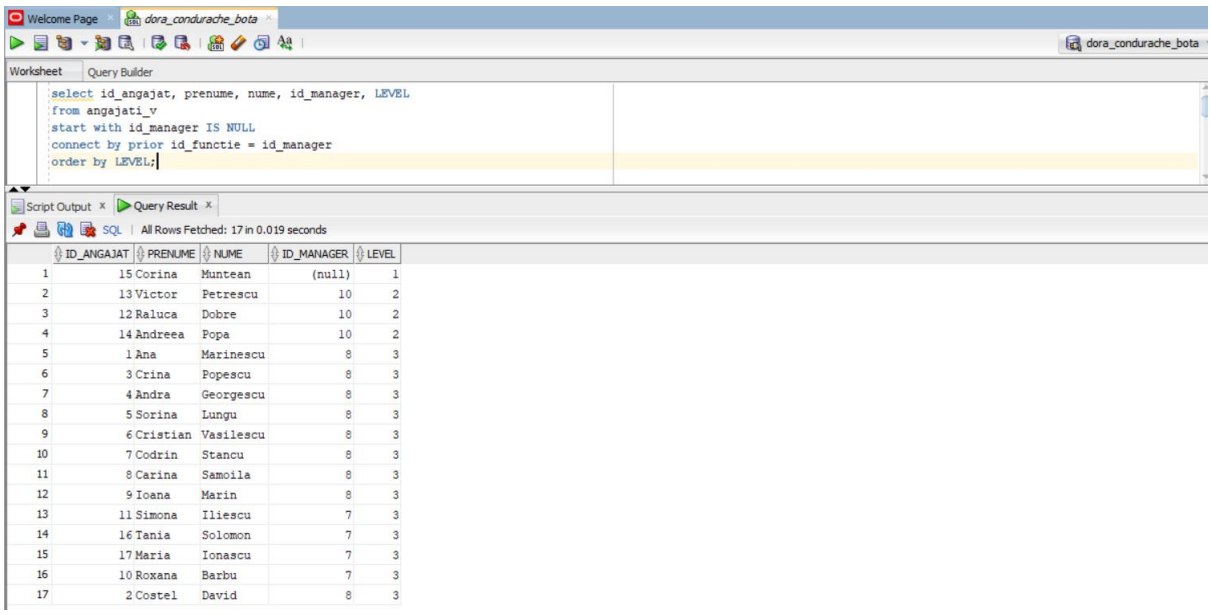


	ID_ANGAJAT	PRENUME	NUME	TELEFON	EMAIL	DATA_ANGAJARE	SALARIUL	ID_FUNCTIE	ID_MANAGER
1	1	Ana	Marinescu	0735264387	marinana65@gmail.com	01-FEB-19	6050	1	8
2	2	Costel	David	0747231645	costeldav4@gmail.com	24-JUN-19	5940	1	8
3	3	Crina	Popescu	0754087648	crinapop3@yahoo.com	13-MAR-23	4620	2	8
4	4	Andra	Georgescu	0734123627	anageorg5@gmail.com	02-JUN-23	4730	2	8
5	5	Sorina	Lungu	0734175329	sorina32@yahoo.com	08-APR-23	4620	2	8
6	6	Cristian	Vasilescu	0782598756	cristian6@yahoo.com	19-NOV-22	3700	3	8
7	7	Codrin	Stancu	0798231593	codrin3@gmail.com	15-OCT-22	3800	3	8
8	8	Carina	Samoila	0752839654	carinasam9@gmail.com	30-APR-21	2700	4	8
9	9	Ioana	Marin	0783175324	ioanam8@yahoo.com	29-APR-22	2600	5	8
10	10	Roxana	Barbu	0712742956	roxanab7@gmail.com	20-AUG-21	3500	6	7
11	11	Simona	Iliescu	0728416422	simonail2@yahoo.com	23-SEP-22	3600	6	7
12	12	Raluca	Dobre	0731768239	ralucad8@yahoo.com	16-SEP-20	5300	7	10
13	13	Victor	Petrescu	0782317534	victorp2@gmail.com	14-JUN-21	5600	8	10
14	14	Andreea	Popa	0752142749	andreeap5@gmail.com	21-MAY-20	4900	9	10
15	15	Corina	Muntean	0728739013	corinam6@yahoo.com	17-JAN-19	6000	10	(null)
16	16	Tania	Solomon	0712742361	tanias9@gmail.com	10-SEP-21	3600	6	7
17	17	Maria	Ionascu	0721749056	mariai8@yahoo.com	06-FEB-22	3500	6	7

Acum, să se afișeze angajații și nivelul lor.

select id_angajat, prenume, nume, id_manager, LEVEL
from angajati_v
start with id_manager IS NULL

connect by prior id_functie = id_manager
order by LEVEL;



```
select id_angajat, prenume, nume, id_manager, LEVEL
from angajati_v
start with id_manager IS NULL
connect by prior id_functie = id_manager
order by LEVEL;
```

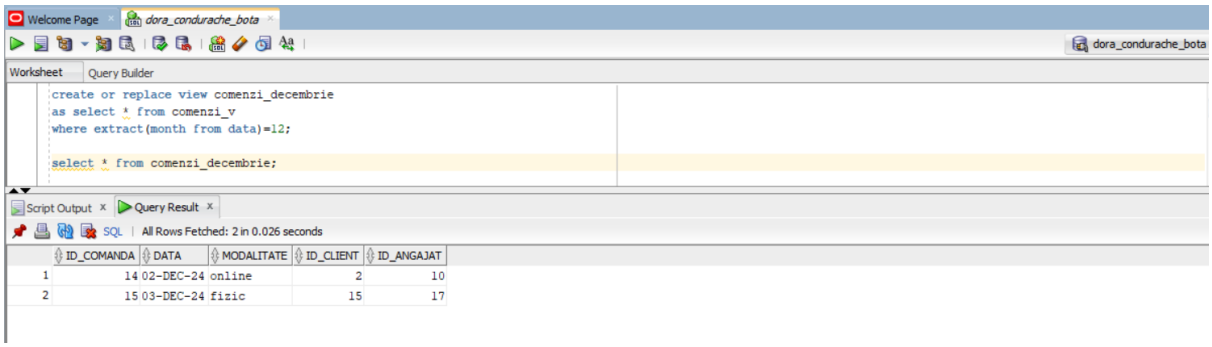
	ID_ANGAJAT	PRENUME	NUME	ID_MANAGER	LEVEL
1	15	Corina	Muntean	(null)	1
2	13	Victor	Petrescu	10	2
3	12	Raluca	Dobre	10	2
4	14	Andreea	Popa	10	2
5	1	Ana	Marinescu	8	3
6	3	Crina	Popescu	8	3
7	4	Andra	Georgescu	8	3
8	5	Sorina	Lungu	8	3
9	6	Cristian	Vasilescu	8	3
10	7	Codrin	Stancu	8	3
11	8	Carina	Samoila	8	3
12	9	Ioana	Marin	8	3
13	11	Simona	Iliescu	7	3
14	16	Tania	Solomon	7	3
15	17	Maria	Ionascu	7	3
16	10	Roxana	Barbu	7	3
17	2	Costel	David	8	3

9. Gestiunea altor obiecte ale bazei de date: vederi, indecsi, sinonime, secvente.

1) Creați un view care să arate comenzile făcute în luna decembrie.

create or replace view comenzi_decembrie
as select * from comenzi_v
where extract(month from data)=12;

select * from comenzi_decembrie;



```
create or replace view comenzi_decembrie
as select * from comenzi_v
where extract(month from data)=12;

select * from comenzi_decembrie;
```

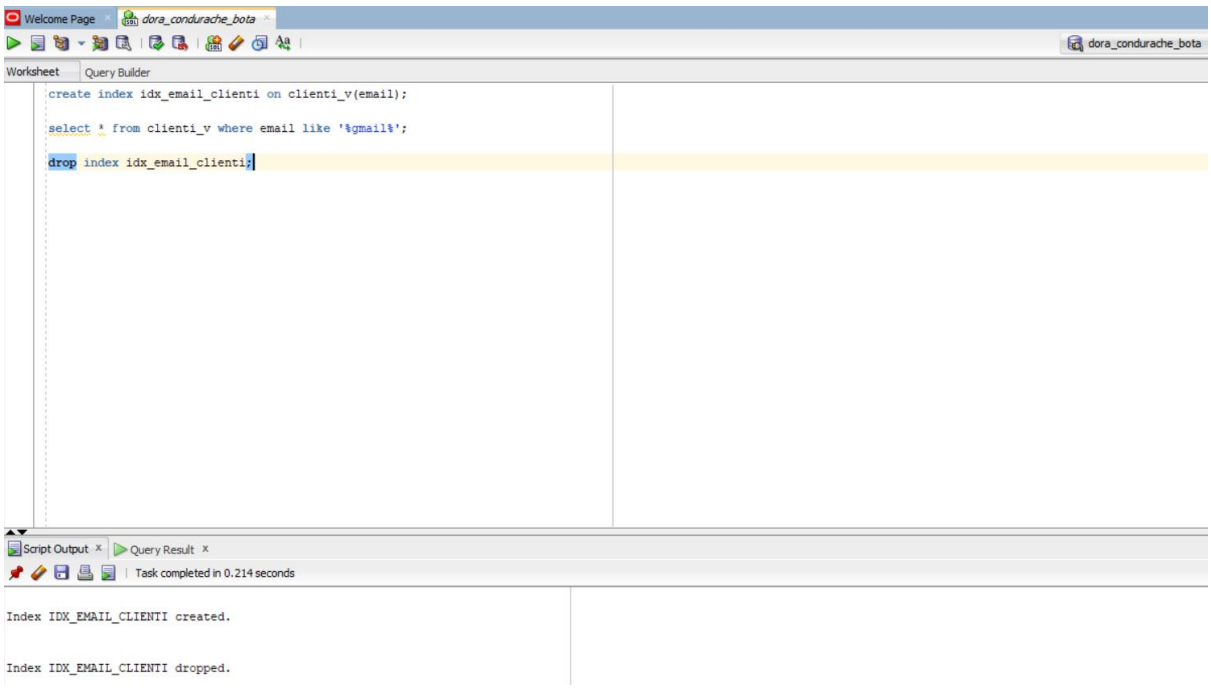
	ID_COMANDA	DATA	MODALITATE	ID_CLIENT	ID_ANGAJAT
1	14	02-DEC-24	online	2	10
2	15	03-DEC-24	fizic	15	17

2) Creați un index pentru email în tabela clienti_v. Stergeți apoi indexul.

create index idx_email_clienti on clienti_v(email);

select * from clienti_v where email like '%gmail%';

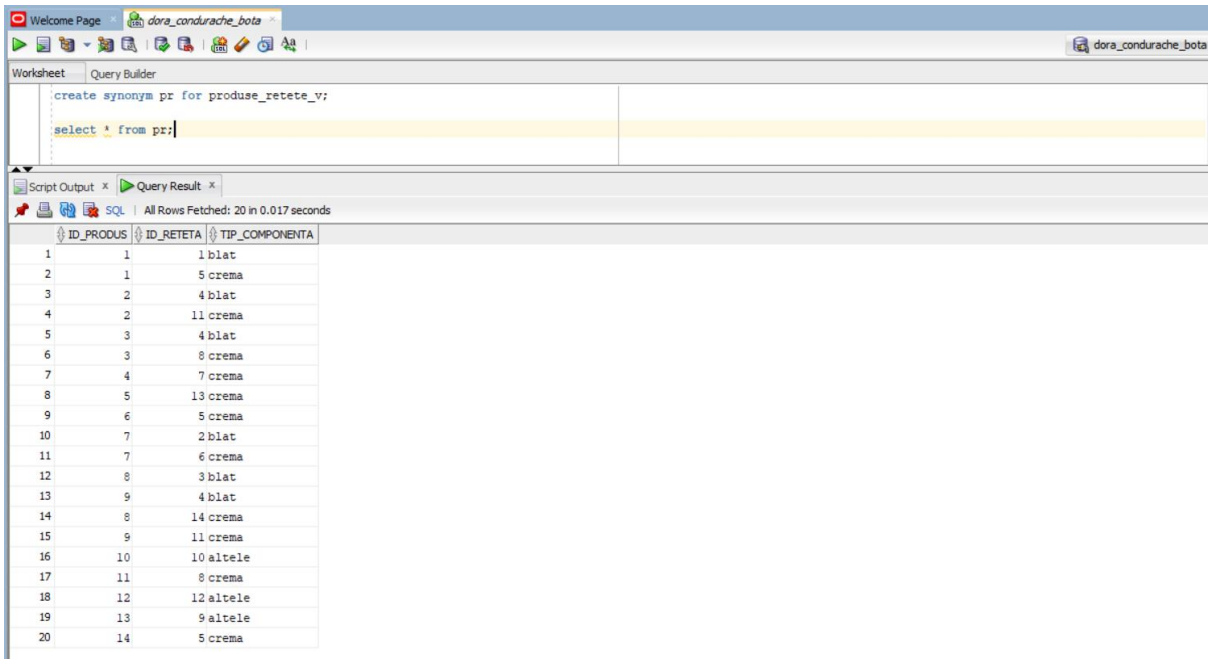
drop index idx_email_clienti;



3) Creați un sinonim pentru tabela produse_retete_v.

create synonym pr for produse_retete_v;

select * from pr;

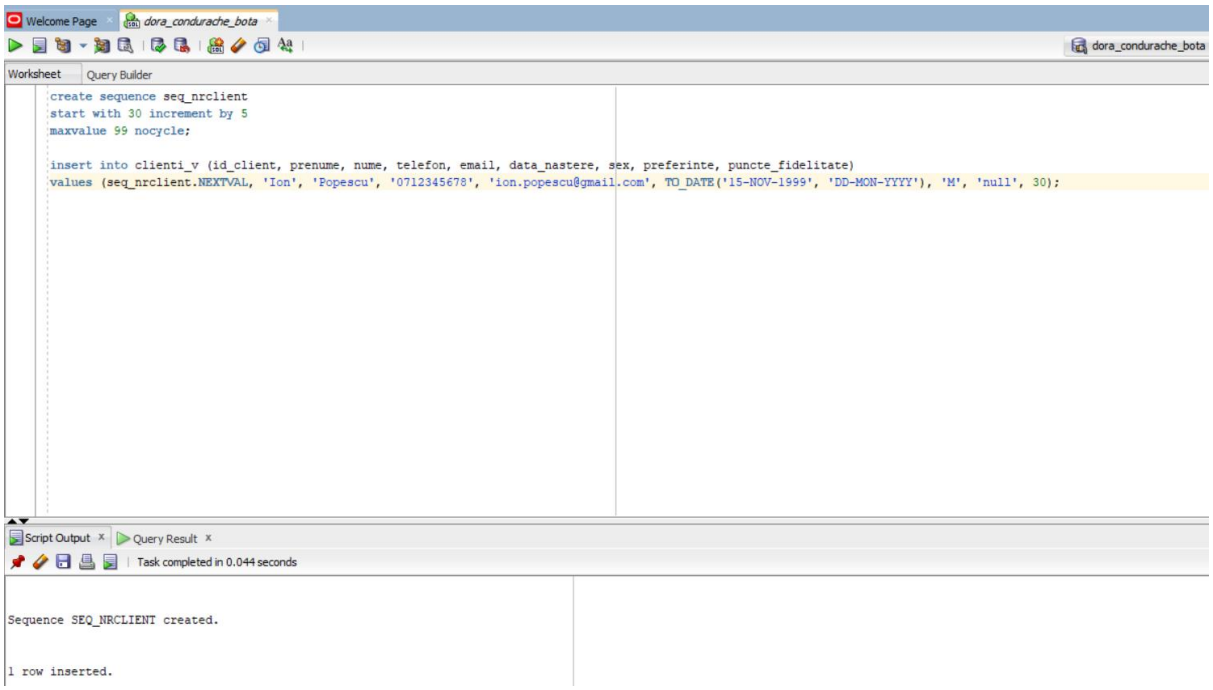


	ID_PRODUS	ID_RETETA	TIP_COMPONENTA
1	1	1	blat
2	1	5	crema
3	2	4	blat
4	2	11	crema
5	3	4	blat
6	3	8	crema
7	4	7	crema
8	5	13	crema
9	6	5	crema
10	7	2	blat
11	7	6	crema
12	8	3	blat
13	9	4	blat
14	8	14	crema
15	9	11	crema
16	10	10	altele
17	11	8	crema
18	12	12	altele
19	13	9	altele
20	14	5	crema

4) Creați o secvență pentru asigurarea unicității cheii primare în tabela client_v.

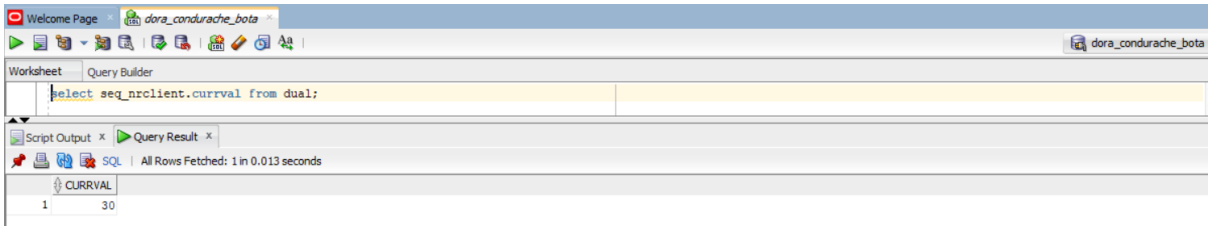
```
create sequence seq_nrclient
start with 30 increment by 5
maxvalue 99 nocycle;

insert into clienti_v (id_client, prenume, nume, telefon, email, data_nastere, sex, preferinte, puncte_fidelitate)
values (seq_nrclient.NEXTVAL, 'Ion', 'Popescu', '0712345678', 'ion.popescu@gmail.com', TO_DATE('15-NOV-1999', 'DD-MON-YYYY'), 'M', 'null', 30);
```



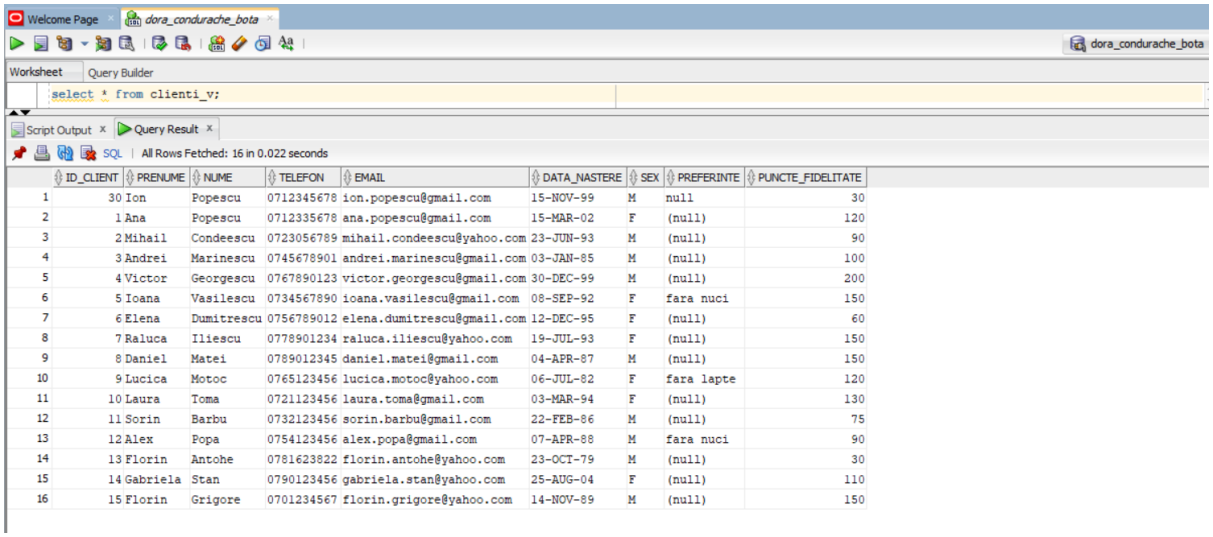
Ca să vedem valoarea curentă a secvenței:

```
select seq_nrclient.currval from dual;
```



Ca să vedem clientul nou adăugat:

```
select * from clienti_v;
```



	ID_CLIENT	PRENUME	NUME	TELEFON	EMAIL	DATA_NASTERE	SEX	PREFERINTE	PUNCTE_FIDELITATE
1	30	Ion	Popescu	0712345678	ion.popescu@gmail.com	15-NOV-99	M	null	30
2	1	Ana	Popescu	0712335678	ana.popescu@gmail.com	15-MAR-02	F	(null)	120
3	2	Mihail	Condeescu	0723056789	mihail.condeescu@yahoo.com	23-JUN-93	M	(null)	90
4	3	Andrei	Marinescu	0745678901	andrei.marinescu@gmail.com	03-JAN-85	M	(null)	100
5	4	Victor	Georgescu	0767890123	victor.georgescu@gmail.com	30-DEC-99	M	(null)	200
6	5	Ioana	Vasilescu	0734567890	ioana.vasilescu@gmail.com	08-SEP-92	F	fara nuci	150
7	6	Elena	Dumitrescu	0756789012	elena.dumitrescu@gmail.com	12-DEC-95	F	(null)	60
8	7	Raluca	Iliescu	0778901234	raluca.iliescu@yahoo.com	19-JUL-93	F	(null)	150
9	8	Daniel	Matei	0789012345	daniel.matei@gmail.com	04-APR-87	M	(null)	150
10	9	Lucica	Motoc	0765123456	lucica.motoc@yahoo.com	06-JUL-82	F	fara lapte	120
11	10	Laura	Toma	0721123456	laura.toma@gmail.com	03-MAR-94	F	(null)	130
12	11	Sorin	Barbu	0732123456	sorin.barbu@gmail.com	22-FEB-86	M	(null)	75
13	12	Alex	Popa	0754123456	alex.popa@gmail.com	07-APR-88	M	fara nuci	90
14	13	Florin	Antohe	0781623822	florin.antohe@yahoo.com	23-OCT-79	M	(null)	30
15	14	Gabriela	Stan	0790123456	gabriela.stan@yahoo.com	25-AUG-04	F	(null)	110
16	15	Florin	Grigore	0701234567	florin.grigore@yahoo.com	14-NOV-89	M	(null)	150